

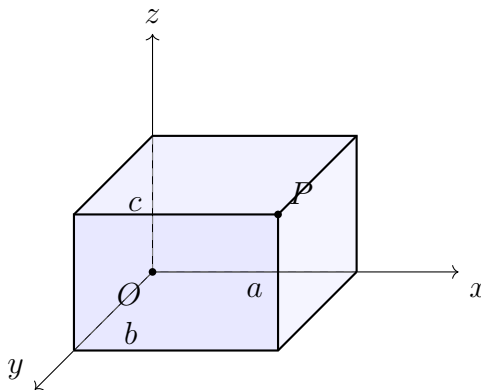
Math 2531 Calculus 3 Review: All Questions
Fall 2026

Exam 1 Material

3D Coordinates and Geometry

1. Distance, Projections, and Spheres

- (a) [2D IN 3D] Only using the standard 2D Pythagorean theorem, find the distance from $(0, 0, 0)$ to $(1, 1, 1)$.
- (b) (Stewart §12.1 #14) Find the distance from the point $(4, -2, 6)$ to each of the following. (a) the xy -plane, (b) the yz -plane, (c) the xz -plane, (d) the x -axis, (e) the y -axis, (f) the z -axis.
- (c) (Stewart §12.1 #4) What are the projections of the point $(2, 3, 5)$ on the xy -, yz -, and xz -planes? Draw a rectangular box with the origin and $(2, 3, 5)$ as opposite vertices and with its faces parallel to the coordinate planes. Label all vertices of the box. Find the length of the diagonal of the box.
- (d) (Stewart §12.1 #9) Find the distance between the points $(3, 5, -2)$ and $(-1, 1, -4)$.
- (e) Find the equation of a sphere if one of its diameters has endpoints $(2, 1, 4)$ and $(4, 3, 10)$.
- (f) (Stewart §12.1 #17) Find an equation of the sphere that passes through the point $(4, 3, -1)$ and has center $(3, 8, 1)$.
- (g) (Stewart §12.1 #16) Find an equation of the sphere with center $(2, -6, 4)$ and radius 5. Describe its intersection with each of the coordinate planes.
- (h) (Stewart §12.1 #20) Show that the equation $x^2 + y^2 + z^2 = 6x - 4y - 10z$ represents a sphere, and find its center and radius.
- (i) The figure shows a rectangular box with one vertex at the origin and the opposite vertex at P , with edges parallel to the coordinate axes.

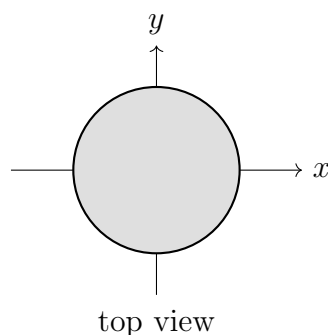
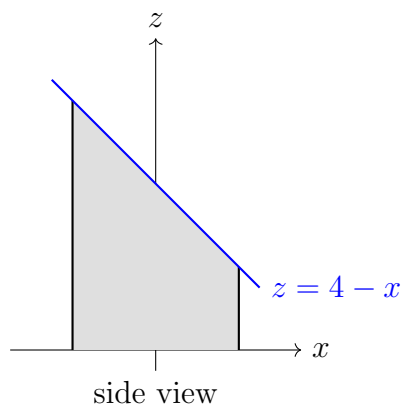


If $|OP| = 7$ and the edges along the x - and y -axes have lengths $a = 3$ and $b = 6$ respectively, find c and the coordinates of P . Then find the lengths of all four space diagonals of the box.

- (j) Find the midpoint of the segment joining $(1, 3, -2)$ and $(5, -1, 4)$.
- (k) Find an equation of the sphere with center $(1, -4, 3)$ and radius 5, and decide whether $(3, -2, 2)$ lies inside, on, or outside it.
- (l) Find the radius of the circle in which the sphere $x^2 + y^2 + z^2 = 9$ meets the plane $z = 1$.

2. Octants, Regions, and Inequalities

- (a) [3D] For what octants is $z \tan(y/x)$ positive or negative?
- (b) (Stewart §12.1 #36) Sketch the region defined by $x^2 + z^2 \leq 25$, $0 \leq y \leq 2$.
- (c) (Stewart §12.1 #27–42) Describe in words the region of $\mathbb{R}^3$ represented by each equation.
- $z = -2$ (#27)
 - $y \geq 1$ (#29)
 - $z = y$ (#32)
 - $x^2 + y^2 = 4$, $z = -1$ (#33)
 - $x^2 + y^2 + z^2 = 4$ (#37)
 - $x^2 + y^2 + z^2 \leq 4$ (#38)
 - $1 \leq x^2 + y^2 \leq 5$ (#40)
 - $0 \leq x \leq 3$, $0 \leq y \leq 3$, $0 \leq z \leq 3$ (#41)
 - $x^2 + y^2 + z^2 > 2z$ (#42)
- (d) (Stewart §12.1 #43) Write inequalities to describe the region between the yz -plane and the vertical plane $x = 5$.
- (e) (Stewart §12.1 #44) Write inequalities to describe the solid cylinder that lies on or below the plane $z = 8$ and on or above the disk in the xy -plane with center the origin and radius 2.
- (f) Sketch the plane given by the equation $2x + 3y + z = 6$ in the first octant. Label the x , y , and z intercepts.
- (g) A solid is bounded below by the disk $x^2 + y^2 \leq 4$ in the xy -plane and above by the slanted plane $z = 4 - x$. The two figures below show the side view (cut by the xz -plane) and the top view.



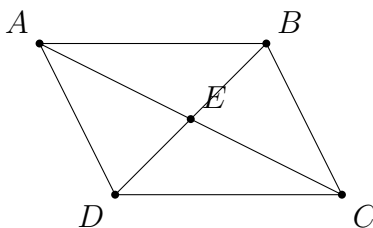
Write a system of inequalities describing this solid, and find the largest and smallest z -values that occur in it.

- (h) Describe in words the solid region $1 \leq x^2 + y^2 + z^2 \leq 4$, and sketch it.
- (i) Sketch the region $x^2 + y^2 \leq z \leq 4$ and describe its bounding surfaces.

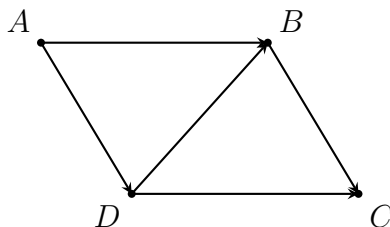
Vectors

3. Vector Pictures and Basics

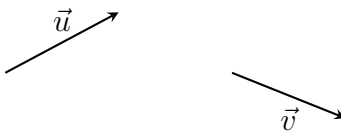
- [1D IN 2D] Find a vector parallel to the line $y = 2x + 1$.
- Parameterize the line $y = 2x - 1$ as a directed curve $\vec{r}(t) = \langle x(t), y(t) \rangle$. Sketch the curve and mark the positions $\vec{r}(0)$ and $\vec{r}(1)$ with arrows indicating the direction of motion.
- A 2D vector has magnitude 2 and has an angle of $\pi/4$ with the positive x -axis. Express the vector in $a\hat{\mathbf{i}} + b\hat{\mathbf{j}}$ form. Also find a unit vector in the same direction.
- (Stewart §12.2 #27) What is the angle between the vector $\hat{\mathbf{i}} + \sqrt{3}\hat{\mathbf{j}}$ and the positive direction of the x -axis?
- (Stewart §12.2 #30) A child pulls a sled through the snow on a level path with a force of 50 N exerted at an angle of 38° above the horizontal. Find the horizontal and vertical components of the force.
- (Stewart §12.2 #9) Find a vector $\vec{a}$ with representation given by the directed line segment $\overrightarrow{AB}$, where $A(-2, 1)$, $B(1, 2)$. Draw $\overrightarrow{AB}$ and the equivalent representation starting at the origin.
- (Stewart §12.2 #19) Let $\vec{a} = \langle -3, 4 \rangle$ and $\vec{b} = \langle 9, -1 \rangle$. Find $\vec{a} + \vec{b}$, $4\vec{a} + 2\vec{b}$, $|\vec{a}|$, and $|\vec{a} - \vec{b}|$.
- (Stewart §12.2 #24) Find a unit vector that has the same direction as $-5\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$.
- Find the unit vector that is in the same direction as the vector from $P(1, 2, 3)$ to $Q(4, 6, 3)$.
- (Stewart §12.2 #44) Let O denote the origin of $\mathbb{R}^n$ and let $A, B \in \mathbb{R}^n$ be points. Let C be the point on segment AB that is twice as close to A as to B (that is, $|AC| = \frac{1}{2}|CB|$). Show that $\vec{OC} = \frac{2}{3}\vec{OA} + \frac{1}{3}\vec{OB}$.
- (Stewart §12.2 #43) If A, B, C are the vertices of a triangle, find $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}$.
- (Stewart §12.2 #41) Find the unit vectors that are parallel to the tangent line to the parabola $y = x^2$ at the point $(2, 4)$.
- (Stewart §12.2 #3) Name all the equal vectors in the parallelogram shown.



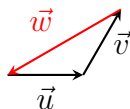
- (Stewart §12.2 #4) Using the vectors in the figure, write each sum or difference as a single vector. (a) $\overrightarrow{AB} + \overrightarrow{BC}$, (b) $\overrightarrow{CD} + \overrightarrow{DB}$, (c) $\overrightarrow{DB} - \overrightarrow{AB}$, (d) $\overrightarrow{DC} + \overrightarrow{CA} + \overrightarrow{AB}$.



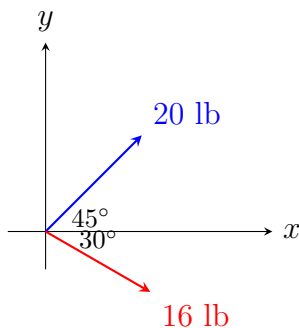
- (o) (Stewart §12.2 #6) Using the vectors $\vec{u}$ and $\vec{v}$ in the figure, sketch each of these. (a) $\vec{u} + \vec{v}$, (b) $\vec{u} - \vec{v}$, (c) $2\vec{u}$, (d) $-\frac{1}{2}\vec{v}$, (e) $3\vec{u} + \vec{v}$, (f) $\vec{v} - 2\vec{u}$.



- (p) (Stewart §12.2 #8) If the vectors in the figure satisfy $|\vec{u}| = |\vec{v}| = 1$ and $\vec{u} + \vec{v} + \vec{w} = \vec{0}$, what is $|\vec{w}|$?



- (q) (Stewart §12.2 #32) Find the magnitude of the resultant force and the angle it makes with the positive x -axis.



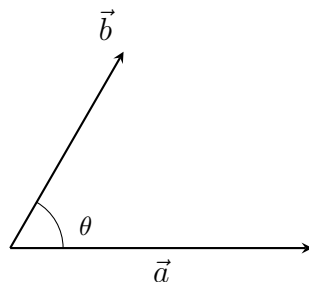
- (r) (Stewart §12.2 #45) Let $\vec{a} = \langle 3, 2 \rangle$, $\vec{b} = \langle 2, -1 \rangle$, $\vec{c} = \langle 7, 1 \rangle$. (a) Draw $\vec{a}$, $\vec{b}$, $\vec{c}$. (b) Show by a sketch that there are scalars s and t such that $\vec{c} = s\vec{a} + t\vec{b}$. (c) Use the sketch to estimate s and t . (d) Find the exact values of s and t .
- (s) Find a vector of magnitude 10 in the direction of $\langle 3, 0, -4 \rangle$.

4. Dot Products and Projections

- (a) [1D IN 2D] If $\vec{a} = \langle 1, 1 \rangle$ and $\vec{b} = \langle -1, 2 \rangle$, draw a picture of and calculate $\text{orth}_{\vec{a}}\vec{b}$ and $\text{proj}_{\vec{a}}\vec{b}$.
- (b) Find the angle θ between the vectors $\vec{a} = \langle 2, -1, 1 \rangle$ and $\vec{b} = \langle 1, 1, 2 \rangle$.
- (c) Determine if the vectors $\vec{u} = \langle 2, 6, -4 \rangle$ and $\vec{v} = \langle -3, 1, 0 \rangle$ are orthogonal, parallel, or neither.
- (d) (Stewart §12.3 #45) Show that $\text{orth}_{\vec{a}}\vec{b} = \vec{b} - \text{proj}_{\vec{a}}\vec{b}$ is orthogonal to $\vec{a}$ (this is why it is called the *orthogonal projection* of $\vec{b}$).
- (e) (Stewart §12.3 #63) **Parallelogram Identity.** The parallelogram identity states that

$$|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2|\vec{a}|^2 + 2|\vec{b}|^2.$$

- (a) Give a geometric interpretation. (b) Prove the identity using the fact $|\vec{x}|^2 = \vec{x} \cdot \vec{x}$.
- (f) In the figure, $|\vec{a}| = 4$ and $|\vec{b}| = 3$, and the angle between them is $\theta = 60^\circ$.



Compute $\vec{a} \cdot \vec{b}$, $\text{proj}_{\vec{a}}\vec{b}$, and the scalar component $\text{comp}_{\vec{a}}\vec{b}$. Mark $\text{proj}_{\vec{a}}\vec{b}$ on the figure.

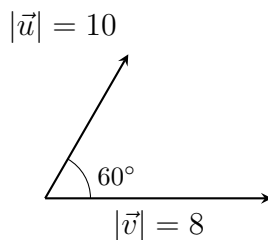
- (g) Find the scalar and vector projections of $\vec{b} = \langle 1, 1, 2 \rangle$ onto $\vec{a} = \langle 2, -1, 2 \rangle$.
- (h) Find the work done by the constant force $\vec{F} = \langle 3, 4, 5 \rangle$ moving an object from $(0, 0, 0)$ to $(2, 1, 0)$.
- (i) Find the direction cosines of $\vec{a} = \langle 1, 2, 2 \rangle$.

5. Cross Products

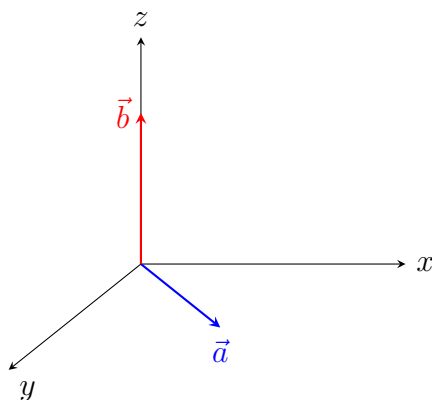
- (a) [2D IN 3D] Determine each of the following without explicit computation. $\hat{\mathbf{i}} \times \hat{\mathbf{j}}$, $\hat{\mathbf{j}} \times \hat{\mathbf{k}}$, $\hat{\mathbf{k}} \times \hat{\mathbf{i}}$, and $-2\hat{\mathbf{k}} \times \hat{\mathbf{j}}$.
- (b) (Stewart §12.4 #9) Find $(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times \hat{\mathbf{k}}$ without using determinants, using only properties of the cross product.
- (c) (Stewart §12.4 #10) Find $\hat{\mathbf{k}} \times (\hat{\mathbf{i}} - 2\hat{\mathbf{j}})$ without using determinants.
- (d) (Stewart §12.4 #11) Find $(\hat{\mathbf{j}} - \hat{\mathbf{k}}) \times (\hat{\mathbf{k}} - \hat{\mathbf{i}})$ without using determinants.
- (e) (Stewart §12.4 #13) State whether each expression is meaningful. If not, explain why. If it is, state whether it is a vector or a scalar. (a) $\vec{a} \cdot (\vec{b} \times \vec{c})$, (b) $\vec{a} \times (\vec{b} \cdot \vec{c})$, (c) $\vec{a} \times (\vec{b} \times \vec{c})$, (d) $\vec{a} \cdot (\vec{b} \cdot \vec{c})$, (e) $(\vec{a} \cdot \vec{b}) \times (\vec{c} \cdot \vec{d})$, (f) $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$.
- (f) Verify that $|\vec{a} \times \vec{b}|$ is the area of the parallelogram described by the 2 vectors.
- (g) Demonstrate that $\vec{a} \cdot (\vec{b} \times \vec{c})$ is the volume of the parallelepiped described by $\vec{a}, \vec{b}, \vec{c}$.
- (h) (Stewart §12.4 #4) Let $\vec{a} = 3\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ and $\vec{b} = 3\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$. Find $\vec{a} \times \vec{b}$ and verify that it is orthogonal to both $\vec{a}$ and $\vec{b}$.
- (i) Find a unit vector orthogonal to both $\vec{u} = \langle 1, 1, 0 \rangle$ and $\vec{v} = \langle 0, 1, 1 \rangle$.
- (j) Let $\vec{a} = \langle 1, 0, 2 \rangle$, $\vec{b} = \langle -1, 3, 1 \rangle$, $\vec{c} = \langle 2, 1, 0 \rangle$.
- Find the volume of the parallelepiped with edge vectors $\vec{a}, \vec{b}, \vec{c}$.
 - The main space diagonal of the parallelepiped runs from the origin to the vertex reached by traveling along all three edges. Find the vector describing this diagonal and compute its length.
- (k) (Stewart §12.4 #27) Find the area of the parallelogram with vertices $A(-3, 0)$, $B(-1, 3)$, $C(5, 2)$, $D(3, -1)$.
- (l) (Stewart §12.4 #29) Let $P(3, 1, 1)$, $Q(5, 2, 4)$, $R(8, 5, 3)$. (a) Find a nonzero vector orthogonal to the plane through P, Q, R . (b) Find the area of triangle PQR .
- (m) (Stewart §12.4 #38) Use the scalar triple product to determine whether the points $A(1, 3, 2)$, $B(3, -1, 6)$, $C(5, 2, 0)$, $D(3, 6, -4)$ lie in the same plane.
- (n) (Stewart §12.4 #45a) Let P be a point not on the line L through points Q and R . Show that the distance from P to L is

$$d = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}|}, \quad \text{where } \vec{a} = \overrightarrow{QR}, \vec{b} = \overrightarrow{QP}.$$

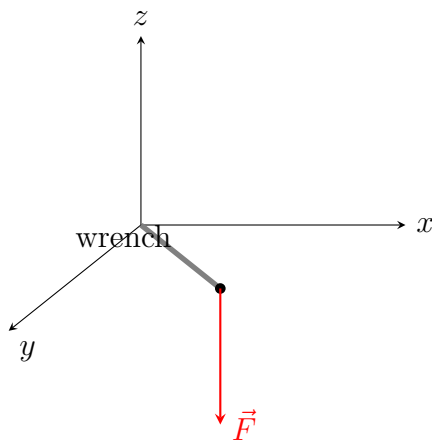
- (o) (Stewart §12.4 #14) Find $|\vec{u} \times \vec{v}|$ and determine whether $\vec{u} \times \vec{v}$ is directed into the page or out of the page.



- (p) (Stewart §12.4 #16) The figure shows a vector $\vec{a}$ in the xy -plane and a vector $\vec{b}$ in the direction of $\hat{\mathbf{k}}$. Their lengths are $|\vec{a}| = 3$ and $|\vec{b}| = 2$.



- (a) Find $|\vec{a} \times \vec{b}|$. (b) Use the right-hand rule to decide whether each component of $\vec{a} \times \vec{b}$ is positive, negative, or zero.
- (q) A torque wrench grips a bolt at the origin. The wrench is 30 cm long, lies in the xy -plane, and points along $\langle 1, 1, 0 \rangle / \sqrt{2}$. A downward force of magnitude 80 N is applied at the far end.



Compute the torque vector $\vec{\tau} = \vec{r} \times \vec{F}$, its magnitude, and its direction.

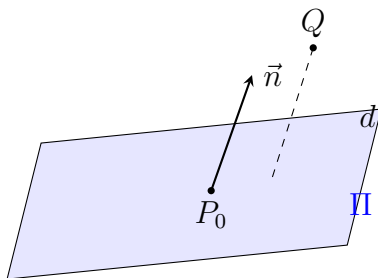
Lines and Planes

6. Lines in Space

- (a) [1D IN 3D] (Stewart §12.5 #1) Determine whether each statement is true or false in $\mathbb{R}^3$.
- i. Two lines parallel to a third line are parallel.
 - ii. Two lines perpendicular to a third line are parallel.
 - iii. Two planes parallel to a third plane are parallel.
 - iv. Two planes perpendicular to a third plane are parallel.
 - v. Two lines parallel to a plane are parallel.
 - vi. Two lines perpendicular to a plane are parallel.
 - vii. Two planes parallel to a line are parallel.
 - viii. Two planes perpendicular to a line are parallel.
 - ix. Two planes either intersect or are parallel.
 - x. Two lines either intersect or are parallel.
 - xi. A plane and a line either intersect or are parallel.
- (b) (Stewart §12.5 #2) Find a vector equation and parametric equations for the line through $(4, 2, -3)$ and parallel to $2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 6\hat{\mathbf{k}}$.
- (c) (Stewart §12.5 #4) Find a vector equation and parametric equations for the line through $(6, 0, -2)$ and parallel to the line $x = 4 - 3t$, $y = -1 + 4t$, $z = 6 + 5t$.
- (d) (Stewart §12.5 #10) Find parametric and symmetric equations for the line through $(2, 1, 0)$ and perpendicular to both $\hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\hat{\mathbf{j}} + \hat{\mathbf{k}}$.
- (e) The parametric equations $x = 3 + 2t$, $y = -1 - t$, $z = 4 + 3t$ describe a line. Write this line in vector form $\vec{r}(t) = \vec{r}_0 + t\vec{v}$, identifying the initial point $\vec{r}_0$ and direction vector $\vec{v}$.
- (f) The symmetric equations of a line are given in the non-standard form $\frac{x-1}{2} = 3(y+2) = \frac{z}{-1}$. Identify a point on the line and a direction vector, then rewrite in standard symmetric form.
- (g) (Stewart §12.5 #5) Find parametric and symmetric equations for the line through $(5, 7, 1)$ that is perpendicular to the plane $3x - 2y + 2z = 8$.
- (h) Find the distance from the point $P(1, 0, -1)$ to the line given by the parametric equations $x = 2 + t$, $y = 1 - 2t$, $z = 3t$.
- (i) Find parametric and symmetric equations of the line through $(2, 1, 0)$ and $(1, 3, -2)$.
- (j) Decide whether the lines $\vec{r}_1 = \langle 1 + t, 2t, 3 - t \rangle$ and $\vec{r}_2 = \langle 2 - s, 3 + s, 1 + 2s \rangle$ are parallel, intersecting, or skew.

7. Planes

- (a) [2D IN 3D] (Stewart §12.5 #24) Find an equation of the plane through $(-3, 4, 2)$ with normal vector $\langle 6, 1, -1 \rangle$.
- (b) (Stewart §12.5 #26) Find an equation of the plane through the origin and perpendicular to the line $x = 1 - 8t$, $y = -1 - 7t$, $z = 4 + 2t$.
- (c) Find the equation of the plane through the origin that is parallel to the vectors $\vec{u} = \langle 1, -1, 2 \rangle$ and $\vec{v} = \langle 2, 1, -1 \rangle$.
- (d) Find the distance from the point $Q(4, 1, -2)$ to the plane defined by the equation $2x - 3y + z = 10$.
- (e) Find the point of intersection of the line $x = 1 + t$, $y = 2 - t$, $z = 3t$ and the plane $2x + y - z = 5$.
- (f) (Stewart §12.5 #57) For the planes $x + y + z = 1$ and $x + 2y + 2z = 1$, do the following.
 - i. Find parametric equations for the line of intersection.
 - ii. Find the angle, in degrees rounded to one decimal place, between the planes.
- (g) (Stewart §12.5 #44) Use intercepts to help sketch the plane $6x + 5y - 3z = 15$.
- (h) The figure shows a plane Π with normal $\vec{n}$ and a point Q not on Π . Use the picture to derive the formula for the distance from Q to Π in terms of $\vec{n}$ and any point $P_0 \in \Pi$.



- (i) Find an equation of the plane through $(1, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 3)$.
- (j) Decide whether the planes $2x - 3y + z = 4$ and $-4x + 6y - 2z = 1$ are parallel, and if so find the distance between them.

8. Quadric Surfaces (Graphing)

- (a) [2D IN 3D] Sketch the paraboloid surface $z = x^2 + y^2$. Sketch the curve formed by the intersection of this surface and the vertical plane $y = x$.
- (b) Sketch the quadric surface defined by the equation $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 1$.
- (c) Sketch the surface $x^2 + y^2 - z^2 = 1$. Identify and label the traces in the xy -plane and the xz -plane.
- (d) For the quadric surface $\frac{x^2}{9} - \frac{y^2}{4} + \frac{z^2}{1} = 1$, do the following.
 - i. Find the traces in all three coordinate planes ($z = 0$, $y = 0$, $x = 0$) and identify each as a conic section (ellipse, hyperbola, or parabola).
 - ii. Use the traces to identify and sketch the surface.
- (e) For each surface below, find the traces parallel to each coordinate plane, identify the type of quadric, and sketch it.
 - i. $4x^2 + y^2 + z^2 = 16$
 - ii. $z = 4x^2 + y^2$
- (f) Sketch the surface given by $z = y^2 - x^2$.
- (g) Sketch the graph of the function $f(x, y) = \sqrt{1 - x^2 - y^2}$. Label at least 3 significant points.
- (h) Sketch the surface $z = \sqrt{x^2 + y^2}$ and identify its intersection with the plane $z = 2$.
- (i) Name each surface: $z^2 = x^2 + y^2$, $x^2 + y^2 - z^2 = 1$, and $4x^2 - y^2 + 2z^2 = 4$.
- (j) Reduce $x^2 + 4y^2 - z^2 + 2x = 3$ by completing the square and name the quadric.

Exam 2 Material

Space Curves and Vector Functions

9. Graphing and Parameterization

(a) [1D IN 3D] (Stewart §13.1 #1) Find the domain of $\vec{r}(t) = \left\langle \ln(t+1), \frac{t}{\sqrt{9-t^2}}, 2^t \right\rangle$.

(b) Let $\vec{r}(t) = \langle \ln(t-1), \arctan(t), \sqrt{4-t^2} \rangle$.

i. Find the domain of $\vec{r}(t)$.

ii. Sketch the projection of the curve onto the xy -plane for t in the domain.

iii. Indicate the direction of increasing t with arrows on your sketch.

(c) (Stewart §13.1 #4) Evaluate

$$\lim_{t \rightarrow 1} \left\langle \frac{t^2 - t}{t - 1}, \sqrt{t + 8}, \frac{\sin(\pi t)}{\ln t} \right\rangle.$$

(d) Sketch the directed curve $\vec{r}(t) = \langle 3 \cos(t), 3 \sin(t), 2 \rangle$ for $0 \leq t \leq 2\pi$. Indicate the direction of motion with an arrow.

(e) (Stewart §13.1 #8) Sketch the curve $\vec{r}(t) = \langle t^2 - 1, t \rangle$ and indicate with an arrow the direction in which t increases.

(f) (Stewart §13.1 #9) Sketch the curve $\vec{r}(t) = \langle 3 \sin t, 2 \cos t \rangle$ and indicate the direction in which t increases.

(g) (Stewart §13.1 #10) Sketch the curve $\vec{r}(t) = e^t \hat{\mathbf{i}} + e^{-t} \hat{\mathbf{j}}$ for $t \in \mathbb{R}$.

(h) (Stewart §13.1 #11) Sketch the curve $\vec{r}(t) = \langle t, 2 - t, 2t \rangle$.

(i) (Stewart §13.1 #12) Sketch $\vec{r}(t) = \langle \sin(\pi t), t, \cos(\pi t) \rangle$.

(j) (Stewart §13.1 #13) Sketch $\vec{r}(t) = \langle 3, t, 2 - t^2 \rangle$.

(k) Sketch the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = y + 1$. Provide a vector-valued function $\vec{r}(t)$ that represents this curve.

(l) Graph the intersection of the sphere $x^2 + y^2 + z^2 = 4$ and the cylinder $(x-1)^2 + y^2 = 1$. Find a parameterization $\vec{r}(t)$ for this curve.

(m) Consider the vector function $\vec{r}(t) = \langle e^t \cos t, e^t \sin t \rangle$.

i. State the domain of $\vec{r}(t)$.

ii. Sketch the directed curve for $0 \leq t \leq 2\pi$, marking the points $\vec{r}(0)$, $\vec{r}(\pi/2)$, $\vec{r}(\pi)$.

What happens to $|\vec{r}(t)|$ as t increases?

(n) (Stewart §13.1 #25–29) For each set of parametric equations below, sketch the curve by hand and identify a simple surface (cone, cylinder, sphere, paraboloid, etc.) that contains it.

i. $x = t \cos t, y = t, z = t \sin t, t \geq 0$ (#25)

ii. $x = \cos t, y = \sin t, z = 1/(1+t^2)$ (#26)

iii. $x = t, y = 1/(1+t^2), z = t^2$ (#27)

iv. $x = \cos t, y = \sin t, z = \cos 2t$ (#28)

v. $x = \cos 8t$, $y = \sin 8t$, $z = e^{0.8t}$, $t \geq 0$ (#29)

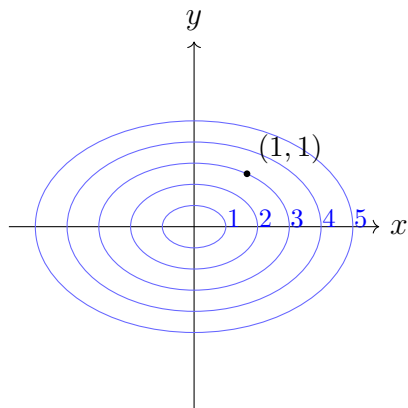
(o) (Stewart §13.1 #39) At what points does the curve $\vec{r}(t) = t\hat{\mathbf{i}} + (2t - t^2)\hat{\mathbf{k}}$ intersect the paraboloid $z = x^2 + y^2$?

(p) (Stewart §13.1 #58) Two particles travel along the curves

$$\vec{r}_1(t) = \langle t, t^2, t^3 \rangle, \quad \vec{r}_2(t) = \langle 1 + 2t, 1 + 6t, 1 + 14t \rangle.$$

Do the particles collide? Do their paths intersect?

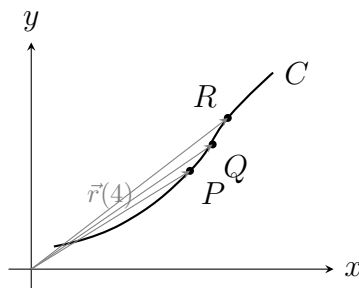
(q) The diagram below shows several level curves of an unknown function $f(x, y)$. Use it to estimate (i) the value of f at $(1, 1)$, (ii) the direction in which f increases fastest from $(1, 1)$, and (iii) whether $\partial f / \partial x$ at $(1, 1)$ is positive, negative, or zero.



(r) Name and sketch the curve $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

10. Derivatives, Integrals, Velocity and Acceleration

- (a) [1D IN 3D] Given the position vector $\vec{r}(t) = \langle t^2, \sin(t), e^t \rangle$, find the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ at $t = 0$.
- (b) (Stewart §13.2 #1) The figure shows a curve C given by a vector function $\vec{r}(t)$, with the tip of $\vec{r}(4)$ at P , the tip of $\vec{r}(4.2)$ at Q , and the tip of $\vec{r}(4.5)$ at R .



- (a) Draw $\vec{r}(4.5) - \vec{r}(4)$ and $\vec{r}(4.2) - \vec{r}(4)$. (b) Draw $\frac{\vec{r}(4.5) - \vec{r}(4)}{0.5}$ and $\frac{\vec{r}(4.2) - \vec{r}(4)}{0.2}$. (c) Write expressions for $\vec{r}'(4)$ and the unit tangent $\vec{T}(4)$. (d) Draw $\vec{T}(4)$.
- (c) (Stewart §13.2 #3) For $\vec{r}(t) = \langle t - 2, t^2 + 1 \rangle$ at $t = -1$, do three things. (a) Sketch the plane curve. (b) Find $\vec{r}'(t)$. (c) Sketch the position vector and tangent vector at the given t .
- (d) (Stewart §13.2 #7) For $\vec{r}(t) = 4 \sin t \hat{i} - 2 \cos t \hat{j}$ at $t = 3\pi/4$, do three things. (a) Sketch the curve. (b) Find $\vec{r}'(t)$. (c) Sketch position and tangent.
- (e) (Stewart §13.2 #39) Evaluate

$$\int_0^1 \left(\frac{1}{t+1} \hat{i} + \frac{1}{t^2+1} \hat{j} + \frac{t}{t^2+1} \hat{k} \right) dt.$$

- (f) (Stewart §13.2 #43) Find $\vec{r}(t)$ if $\vec{r}'(t) = 2t \hat{i} + 3t^2 \hat{j} + \sqrt{t} \hat{k}$ and $\vec{r}(1) = \hat{i} + \hat{j}$.
- (g) Find the unit tangent vector $\vec{T}(t)$ and the unit normal vector $\vec{N}(t)$ for the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$.
- (h) Find the tangential and normal components of acceleration (a_T and a_N) for the curve $\vec{r}(t) = \langle t, t^2, 3t \rangle$.
- (i) (Stewart §13.4 #3) Find the velocity, acceleration, and speed of a particle with position function $\vec{r}(t) = \langle -\frac{1}{2}t^2, t \rangle$ at $t = 2$. Sketch the path and draw the velocity and acceleration vectors at $t = 2$.
- (j) (Stewart §13.4 #10) Find the velocity, acceleration, and speed of a particle with position function $\vec{r}(t) = \langle 2 \cos t, 3t, 2 \sin t \rangle$.
- (k) (Stewart §13.4 #16) Find the velocity and position vectors of a particle with $\vec{a}(t) = \sin t \hat{i} + 2 \cos t \hat{j} + 6t \hat{k}$, $\vec{v}(0) = -\hat{k}$, $\vec{r}(0) = \hat{j} - 4\hat{k}$.
- (l) (Stewart §13.4 #35) A particle has position function $\vec{r}(t)$. If $\vec{r}'(t) = \vec{c} \times \vec{r}(t)$ where $\vec{c}$ is a constant vector, describe the path of the particle.
- (m) (Stewart §13.4 #36) (a) If a particle moves along a straight line, what can you say about its acceleration vector? (b) If a particle moves with constant speed along a curve, what can you say about its acceleration vector?

- (n) (Stewart §13.4 #37) Find the tangential and normal components of the acceleration vector for $\vec{r}(t) = (t^2 + 1)\hat{\mathbf{i}} + t^3\hat{\mathbf{j}}$, $t \geq 0$.
- (o) If a particle moves with constant speed, prove that the velocity and acceleration vectors are always orthogonal.
- (p) A projectile is fired with an initial speed of 500 m/s at an angle of 30 degrees. Find the range of the projectile.
- (q) An electron moves through a magnetic field with a velocity vector at time t in nanoseconds given by

$$\vec{v}(t) = \langle -\sin(t), \frac{1}{\sqrt{2}}\cos(t), \frac{1}{\sqrt{2}}\cos(t) \rangle$$

At time $t = 0$, the electron is located at the initial position $\vec{r}(0) = \langle 1, 0, 0 \rangle$.

- i. Determine the position vector $\vec{r}(t)$ for all $t \geq 0$.
 - ii. Show that the electron is trapped on the surface of a sphere centered at the origin by finding the equation of that sphere.
- (r) Find parametric equations of the tangent line to $\vec{r}(t) = \langle t, t^2, t^3 \rangle$ at $t = 1$.
 - (s) A projectile has acceleration $\langle 0, 0, -g \rangle$, initial velocity $\langle v_0, 0, 0 \rangle$, and starts at the origin. Find its position $\vec{r}(t)$.

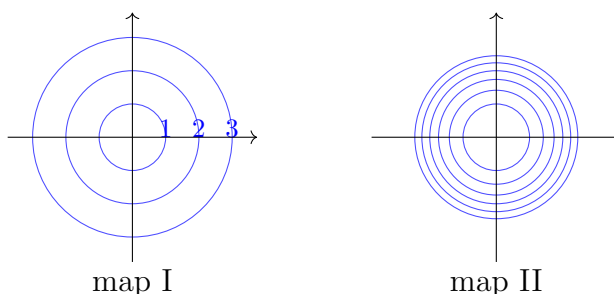
11. Arc Length and Curvature

- (a) [1D IN 3D] (Stewart §13.3 #1) For $\vec{r}(t) = \langle 3 - t, 2t, 4t + 1 \rangle$, $1 \leq t \leq 3$, do two things.
(a) Use the arc length integral to compute the length of the segment. (b) Compute the length using the distance formula and compare.
- (b) Find the length of the arc of the circular helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$ from $t = 0$ to $t = 2\pi$.
- (c) (Stewart §13.3 #21) For $\vec{r}(t) = \langle t, t^2, 4 \rangle$, find $\vec{T}(t)$, $\vec{N}(t)$, and $\kappa(t)$.
- (d) Calculate the curvature $\kappa(t)$ for the space curve $\vec{r}(t) = \langle t, t^2, \frac{2}{3}t^3 \rangle$ using the formula $\kappa = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$.
- (e) Find the point on the curve $y = e^x$ where the curvature is maximized.
- (f) Consider the ellipse $\frac{x^2}{4} + y^2 = 1$.
- Compute the curvature κ at the points $(2, 0)$ and $(0, 1)$.
 - For each point, find the radius of curvature $\rho = 1/\kappa$ and the center of the osculating circle. (The center lies at distance ρ from the point in the direction of $\vec{N}$.)
 - Sketch the ellipse and both osculating circles. At which point is the ellipse curving more sharply?
- (g) Find the length of $\vec{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$ for $0 \leq t \leq 1$.
- (h) Find the curvature of $y = x^2$ at the origin and at $(1, 1)$.
- (i) Find the unit tangent $\vec{T}$ and unit normal $\vec{N}$ for $\vec{r}(t) = \langle \cos t, \sin t \rangle$.
- (j) Find the radius of the osculating circle of $y = \ln x$ at $(1, 0)$.

Functions of Multiple Variables

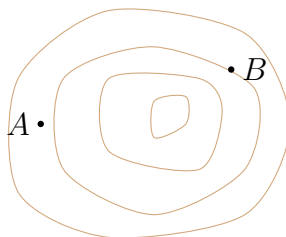
12. Domains and Level Sets

- (a) [2D IN 3D] Sketch the level curves of $f(x, y) = x^2 + 4y^2$ for the values $k = 1, k = 4, k = 9$.
- (b) Sketch the domain of the function $f(x, y) = \ln(y - x^2)$.
- (c) Describe the level surfaces of $f(x, y, z) = x^2 - y^2 - z^2$ for $k > 0$, $k = 0$, and $k < 0$.
- (d) Sketch the level curves for $f(x, y) = \frac{x}{x^2 + y^2}$. Show that these curves are all circles.
- (e) Sketch three level surfaces of the function $f(x, y, z) = x^2 + y^2 + z^2$.
- (f) (Stewart §14.1 #36) Two contour maps below are shown. One belongs to a function whose graph is a cone, the other to a function whose graph is a paraboloid. Decide which is which and justify your answer.



(Level sets in map I are evenly spaced. In map II they bunch up as the radius grows.)

- (g) (Stewart §14.1 #37, adapted) The topographic map below shows a feature called “Lonesome Mountain.” Consider the marked points A (on the western flank, where contours are closely spaced) and B (between two rings on the eastern ridge). Describe in words the shape of the terrain near each.



- (h) Find and sketch the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$ and of $g(x, y) = \ln(x - y)$.
- (i) Sketch several level curves of $f(x, y) = y - x^2$ and of $g(x, y) = x^2 - y^2$.

13. Limits and Continuity

- (a) [2D] Evaluate the limit, or show it does not exist, for $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$.
- (b) Use polar coordinates to evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2}$.

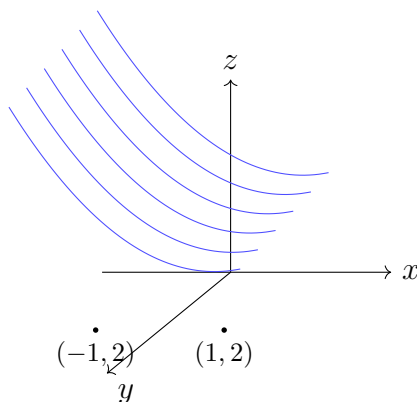
- (c) Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^3}{x^2 + y^6}$ does not exist, even though the limit is 0 along every straight line through the origin.
- (d) Determine the set of points at which $f(x, y) = \frac{1}{1-x^2-y^2}$ is continuous.
- (e) Show that the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{5y^4 \cos^2 x}{x^4 + y^4}$ does not exist.
- (f) (Stewart §14.2 #49) Let

$$f(x, y) = \begin{cases} \frac{x^2 y^3}{2x^2 + y^2} & (x, y) \neq (0, 0) \\ 1 & (x, y) = (0, 0) \end{cases}.$$

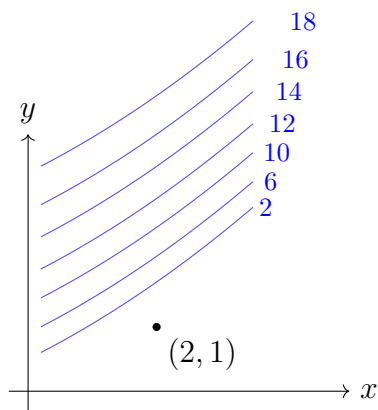
Is f continuous at the origin? Justify carefully.

14. Basic Partial Derivatives Practice

- (a) [1D IN 3D] Find f_x and f_y for $f(x, y) = x^4y^2 - 10x^2y + 7y - 3$.
- (b) Find f_x and f_y for $f(x, y) = \sin(x^2y + y^3)$.
- (c) Find f_x and f_y for $f(x, y) = e^{xy} \ln(x^2 + 1)$.
- (d) Find f_x and f_y for $f(x, y) = \frac{x+y}{x-y}$.
- (e) Find f_x and f_y for $f(x, y) = \arctan(y/x)$.
- (f) For $f(x, y) = e^x \sin(y)$, verify that $f_{xy} = f_{yx}$.
- (g) (Stewart §14.3 #13) Find the first partial derivatives of $z = \ln(x + t^2)$ (treating x and t as the two independent variables).
- (h) (Stewart §14.3 #14) Find the first partial derivatives of $w = u/v^2$.
- (i) (Stewart §14.3 #16) Find the first partials of $g(x, y) = (x + xy^2)^3$.
- (j) (Stewart §14.3 #19) Find the first partials of $f(x, y) = \frac{ax + by}{cx + dy}$.
- (k) (Stewart §14.3 #26) Find the first partials of $F(\alpha, \beta) = \int_{\alpha}^{\beta} \sqrt{t^3 + 1} dt$.
- (l) (Stewart §14.3 #37) For $R(s, t) = t e^{s/t}$, find $R_t(0, 1)$.
- (m) (Stewart §14.3 #46) Find $\partial z / \partial x$ and $\partial z / \partial y$ for each of the following.
 - i. $z = f(x) g(y)$
 - ii. $z = f(x + y)$
 - iii. $z = f(xy)$
 - iv. $z = f(x/y)$
- (n) (Stewart §14.3 #4–5) The figure shows the graph of a function $f(x, y)$. The surface near the point $(\pm 1, 2)$ slopes down as x increases (steeply on the left, gently on the right) and slopes slightly down as y increases.



- Determine the sign of each. (a) $f_x(1, 2)$, (b) $f_y(1, 2)$, (c) $f_x(-1, 2)$, (d) $f_y(-1, 2)$.
- (o) (Stewart §14.3 #6) A contour map of a function f is shown below.



Use the contour map to estimate $f_x(2, 1)$ and $f_y(2, 1)$.

- (p) Let $V(r, h) = rh \cos(\frac{\pi}{2}e^{rh})$, where r and h are functions of time t given by

$$r(t) = \cos^2(t), \quad h(t) = \sin(3t)$$

Find $\frac{dV}{dt}|_{t=\pi}$. Consider the curve $\vec{p}(t) = (r(t), h(t), V(r(t), h(t)))$ along the graph of V . What does $\frac{dV}{dt}|_{t=\pi}$ represent in terms of $\vec{p}(t)$?

- (q) Find f_x and f_y for $f(x, y) = \ln(x^2 + y^2)$ and for $g(x, y) = \frac{x}{x + y}$.
- (r) Find all second partial derivatives of $f(x, y) = xe^y + ye^x$.
- (s) For $f(x, y, z) = xyz + x^2z$, find f_x , f_y , and f_z .

15. Implicit Differentiation

- (a) [2D IN 3D] Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $x^2 + y^2 + z^2 = 3xyz$.
- (b) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $xe^y + yz + ze^x = 0$.
- (c) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $\sin(x + y) + \sin(y + z) + \sin(x + z) = 0$.
- (d) Find $\partial z / \partial x$ if $e^z = xyz$.
- (e) Find dy/dx for $x^2 + xy + y^2 = 3$ using the ratio of partials of F .

16. Chain Rule Practice

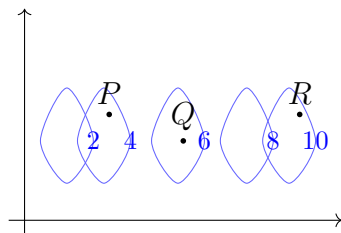
- (a) [1D IN 3D] (Stewart §14.5 #3) Use the chain rule to find dz/dt , where $z = xy^3 - x^2y$, $x = t^2 + 1$, $y = t^2 - 1$.
- (b) (Stewart §14.5 #5) Find dz/dt for $z = \sin x \cos y$, $x = \sqrt{t}$, $y = 1/t$.
- (c) (Stewart §14.5 #6) Find dw/dt for $w = \sqrt{1 + xy}$, $x = \tan t$, $y = \arctan t$.
- (d) (Stewart §14.5 #11) Find $\partial z/\partial s$ and $\partial z/\partial t$ for $z = (x - y)^5$, $x = s^2t$, $y = st^2$.
- (e) (Stewart §14.5 #13) Find $\partial z/\partial s$ and $\partial z/\partial t$ for $z = \ln(3x + 2y)$, $x = s \sin t$, $y = t \cos s$.
- (f) (Stewart §14.5 #15) Find $\partial z/\partial s$ and $\partial z/\partial t$ for $z = (\sin \theta)/r$, $r = st$, $\theta = s^2 + t^2$.
- (g) (Stewart §14.5 #20) Suppose f is a differentiable function of x and y , and $g(r, s) = f(2r - s, s^2 - 4r)$. Using the table $f(1, 2) = 6$, $f_x(1, 2) = 2$, $f_y(1, 2) = 5$, calculate $g_r(1, 2)$ and $g_s(1, 2)$.
- (h) (Stewart §14.5 #22) Use a tree diagram to write out the Chain Rule for the case $w = f(x, y, z)$, where $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$.
- (i) Given $w = f(x, y)$, where $x = s + t$ and $y = s - t$, show that $\frac{\partial w}{\partial s} + \frac{\partial w}{\partial t} = 2 \frac{\partial f}{\partial x}$.
- (j) Let $z = e^{xy} \cos(x + y)$, where $x = st$ and $y = s/t$. Find $\frac{\partial z}{\partial s}$.
- (k) If $u = f(x - at) + g(x + at)$, show that u satisfies the wave equation $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$.
- (l) Let $z = x^2y$, where $x = r \cos \theta$ and $y = r \sin \theta$. Find $\frac{\partial z}{\partial \theta}$ using the chain rule.
- (m) If $z = x^2y$, $x = t^2$, $y = t^3$, find dz/dt two ways and compare.
- (n) If $w = x^2 + y^2 + z^2$ with $x = \cos t$, $y = \sin t$, $z = t$, find dw/dt .
- (o) If $z = e^x \sin y$ with $x = st$ and $y = s^2t$, find $\partial z/\partial s$ and $\partial z/\partial t$.
- (p) The radius of a cone grows at 2 cm/s and its height shrinks at 3 cm/s. Find the rate of change of the volume when $r = 10$ and $h = 20$.

17. Tangent Planes and Linearization

- (a) [2D IN 3D] Find an equation of the tangent plane to the surface $x^2 + 2y^2 + 3z^2 = 21$ at the point $(2, 2, 1)$.
- (b) (Stewart §14.4 #2) Find an equation of the tangent plane to the surface $z = y^2 \sin x$ at the point $(\pi/2, -2, 4)$.
- (c) (Stewart §14.4 #4) Find the tangent plane to $z = (x + 2)^2 - 2(y - 1)^2 - 5$ at $(2, 3, 3)$.
- (d) (Stewart §14.4 #6) Find the tangent plane to $z = y^2 e^x$ at $(0, 3, 9)$.
- (e) Find the linear approximation of $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$ and use it to estimate $f(1.95, 1.08)$.
- (f) (Stewart §14.4 #15) Explain why $f(x, y) = x^3 y^2$ is differentiable at $(-2, 1)$ and find the linearization $L(x, y)$ at that point.
- (g) (Stewart §14.4 #18) Explain why $f(x, y) = \sqrt{xy}$ is differentiable at $(1, 4)$, and find its linearization there.
- (h) (Stewart §14.4 #22) Explain why $f(x, y) = y + \sin(x/y)$ is differentiable at $(0, 3)$, and find its linearization there.
- (i) Sketch the traces $z = f(x, y_0)$ and $z = f(x_0, y)$ and explain how they determine the tangent plane.
- (j) Find the tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$.
- (k) Find the tangent plane to $z = \ln(2x + y)$ at $(-1, 3, 0)$.
- (l) Find dz for $z = x^2 y^3$ and use it to estimate the change as (x, y) moves from $(1, 2)$ to $(1.05, 1.98)$.

18. Directional Derivatives and Gradients

- (a) [2D IN 3D] Find the directional derivative of $f(x, y) = x^2y^3$ at $P(1, 2)$ in the direction of $\vec{v} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$.
- (b) (Stewart §14.6 #7) Find the directional derivative of $f(x, y) = \arctan(xy)$ at $(2, -3)$ in the direction indicated by the angle $\theta = 3\pi/4$.
- (c) (Stewart §14.6 #9) For $f(x, y) = x/y$ at $P(2, 1)$ with unit direction $\vec{u} = \frac{3}{5}\hat{\mathbf{i}} + \frac{4}{5}\hat{\mathbf{j}}$, do three things. (a) Find ∇f . (b) Evaluate ∇f at P . (c) Find the rate of change of f at P in the direction $\vec{u}$.
- (d) (Stewart §14.6 #10) Same parts (a)(b)(c) for $f(x, y) = x^2 \ln y$ at $P(3, 1)$ with $\vec{u} = -\frac{5}{13}\hat{\mathbf{i}} + \frac{12}{13}\hat{\mathbf{j}}$.
- (e) (Stewart §14.6 #14) Find the directional derivative of $f(x, y) = x/(x^2 + y^2)$ at $(1, 2)$ in the direction $\vec{v} = \langle 3, 5 \rangle$.
- (f) (Stewart §14.6 #22) Find the directional derivative of $f(x, y) = x/y^2$ at $P(3, -1)$ in the direction of $Q(-2, 11)$.
- (g) Find the maximum rate of change of $f(x, y, z) = x + y/z$ at the point $(1, 1, 1)$ and the direction in which it occurs.
- (h) (Stewart §14.6 #26) The contour map of a function f is shown below. At each of the points P, Q, R , draw an arrow indicating the direction of the gradient vector.

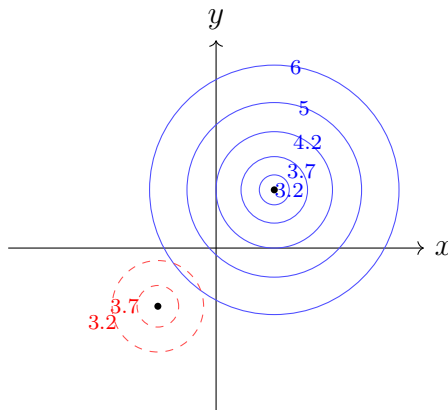


- (i) (Stewart §14.6 #27) Find the maximum rate of change of $f(x, y) = 5xy^2$ at $(3, -2)$ and the direction in which it occurs.
- (j) (Stewart §14.6 #33b) Use the fact that f decreases most rapidly in the direction $-\nabla f$ to find the direction in which $f(x, y) = x^4y - x^2y^3$ decreases fastest at $(2, -3)$. What is the rate of decrease?
- (k) (Stewart §14.6 #38) The temperature at a point (x, y, z) is given by $T(x, y, z) = 200e^{-x^2-3y^2-9z^2}$. (a) Rate of change of T at $P(2, -1, 2)$ in the direction toward $(3, -3, 3)$. (b) In which direction does T increase fastest at P ? (c) What is the maximum rate of increase at P ?
- (l) Show that the gradient vector $\nabla f(x_0, y_0, z_0)$ is always perpendicular to the level surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0) .
- (m) Find all points (x, y) where the directional derivative of $f(x, y) = x^2 + y^2$ in the direction $\vec{u} = \langle 1/\sqrt{2}, 1/\sqrt{2} \rangle$ is zero.
- (n) Sketch the level curves of $f(x, y) = x^2 + y^2$. At the point $(1, 1)$, draw the gradient vector ∇f .
- (o) A solid object has a temperature distribution given by $F(x, y, z) = x^2 + 2y^2 + 3z^2$ (in degrees Celsius).

- i. Show that the point $P(1, 2, 1)$ lies on the isothermal (constant temperature) surface $F(x, y, z) = 12$.
 - ii. Find the gradient vector ∇F at the point P .
 - iii. Find the equation of the tangent plane to this isothermal surface at point P .
 - iv. Heat flows in the direction of the greatest temperature change. At point P , for what unit vector $\hat{u}$ is the magnitude of the directional derivative $|D_{\hat{u}}F|$ the largest?
- (p) Find the maximum rate of change of $f(x, y) = \sin(xy)$ at $(1, 0)$ and the direction in which it occurs.
- (q) At $(1, 2)$, in what direction does $f(x, y) = x^2 + y^2$ decrease fastest, and in what directions is its rate of change zero?

19. Maxima and Minima

- (a) [2D IN 3D] Find and classify the critical points of $f(x, y) = x^4 + y^4 - 4xy + 1$.
- (b) (Stewart §14.7 #3) The level curves of $f(x, y) = 4 + x^3 + y^3 - 3xy$ are shown below, each contour labeled with its f -value. Predict the location of the critical points and whether each is a saddle, local max, or local min. Then use the Second Derivatives Test to confirm.



- (c) Find the absolute maximum and minimum of $f(x, y) = xy$ on the disk $x^2 + y^2 \leq 1$.
- (d) Find the shortest distance from the origin to the plane $x + 2y + 3z = 12$ using the Second Derivative Test.
- (e) For $f(x, y) = x^2 + y^2 - 2x$, use the second derivative test to classify the critical points, then find the absolute maximum and absolute minimum values on the closed disk $x^2 + y^2 \leq 4$.
- (f) Sketch a saddle point surface. Explain how the determinant of the Hessian $D < 0$ captures this behavior.
- (g) Find 3 positive numbers whose sum is 100 and whose product is a maximum.
- (h) Find and classify the critical points of $f(x, y) = x^2 + y^2 - 4x - 6y + 13$.
- (i) [1D IN 2D] Classify the critical points of $g(x) = x^3 - 3x$ with the second derivative test, then do the same for $f(x, y) = x^3 - 3x + y^2$ and explain how the added y^2 turns the maximum into a saddle.

Exam 3 Material

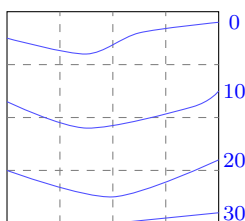
Multiple Integrals

20. The Dome as a Volume (running example)

- (a) [1D] The companion curve is $y = 1 - x^2$ on $[-1, 1]$, the dome's vertical slice through the x -axis. Sketch it and find the area under it, $\int_{-1}^1 (1 - x^2) dx$.
- (b) [2D IN 3D] Sketch the dome over the unit disk and its shadow $x^2 + y^2 \leq 1$. Set up the volume $\iint_D (1 - x^2 - y^2) dA$ with the inner strip in y . Read the limits off the shadow, but do not evaluate yet.
- (c) [2D IN 3D] The same volume in polar is $\int_0^{2\pi} \int_0^1 (1 - r^2) r dr d\theta$. Evaluate it, and say in one line why the extra r appears and why polar is the natural coordinate system for a disk.
- (d) [2D] Reverse the order of your rectangular integral from part (b) and confirm the shadow is described the other way. Which order would you rather carry out, and why?
- (e) [3D] Sweep the solid vertically. Write the volume as $\iiint_E dV$ with $0 \leq z \leq 1 - x^2 - y^2$ over the disk shadow, then redo it in cylindrical as $\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} r dz dr d\theta$. Check it matches part (c).
- (f) [3D] Give the solid density $\delta = 1$. By symmetry $\bar{x} = \bar{y} = 0$. Find $\bar{z} = \frac{1}{V} \iiint_E z dV$ in cylindrical and mark the balance height on your sketch.
- (g) [3D] Cap the same disk with the hemisphere $z = \sqrt{1 - x^2 - y^2}$ instead of the dome. Set up its volume in spherical, $\int_0^{2\pi} \int_0^{\pi/2} \int_0^1 \rho^2 \sin \phi d\rho d\phi d\theta$, and sketch the spherical wedge whose three edges give $\rho^2 \sin \phi$. Compare the two solids over the shared shadow.

21. Double Integrals (Rectangular and General Regions)

- (a) [2D] (Stewart §15.1 #2a) For $R = [0, 4] \times [-1, 2]$ and the integral $\iint_R (1 - xy^2) dA$, use a Riemann sum with $m = 2$, $n = 3$, sampling at the *lower right* corner of each subrectangle.
- (b) (Stewart §15.1 #4b) Use the Midpoint Rule with $m = n = 2$ to estimate the volume under $z = 1 + x^2 + 3y$ above $R = [1, 2] \times [0, 3]$.
- (c) (Stewart §15.1 #9) Evaluate $\iint_R \sqrt{2} dA$ where $R = \{(x, y) : 2 \leq x \leq 6, -1 \leq y \leq 5\}$ by first identifying it as the volume of a solid.
- (d) (Stewart §15.1 #10) Evaluate $\iint_R (2x + 1) dA$ where $R = \{(x, y) : 0 \leq x \leq 2, 0 \leq y \leq 4\}$ by identifying it as a volume.
- (e) (Stewart §15.1 #14) For $f(x, y) = y\sqrt{x+2}$, find $\int_0^2 f(x, y) dx$ and $\int_0^3 f(x, y) dy$.
- (f) (Stewart §15.1 #7) A contour map of a function f on the square $R = [0, 4] \times [0, 4]$ is shown below, each blue curve labeled with the corresponding value of f . Use the Midpoint Rule with $m = n = 2$ to estimate $\iint_R f dA$, and estimate the average value of f .



- (g) Evaluate the double integral $\iint_R (x^2y + 4) dA$, where $R = [0, 3] \times [1, 2]$.
- (h) Sketch the region of integration and evaluate $\int_0^1 \int_x^1 e^{y^2} dy dx$ by reversing the order of integration.
- (i) (Stewart §15.2 #7) For $f(x, y) = 2y$ and the region D bounded by $y = x$ and $y = 3x - x^2$, which meet at the origin and at $(2, 2)$, do two things. (a) Express $\iint_D f dA$ as an iterated integral. (b) Evaluate it.
- (j) (Stewart §15.2 #11) Evaluate $\iint_D \frac{y}{x^2 + 1} dA$, where $D = \{(x, y) : 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$.
- (k) (Stewart §15.2 #8) For $f(x, y) = x + y$, and D the region bounded by $y = x - 2$, $y = \sqrt{x}$, and $y = 6 - x$, do two things. (a) Write the integral. (b) Evaluate it.
- (l) Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy -plane bounded by $y = x^2$ and $x = y^2$.
- (m) Calculate $\iint_D (x + y) dA$, where D is the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$.
- (n) Find the area of the region D bounded by $y = x$ and $y = x^2 - 2x$.
- (o) Find the volume under $z = 4 - x - y$ over $R = [0, 1] \times [0, 2]$.
- (p) Evaluate $\iint_D x dA$ where D is bounded by $y = x$ and $y = x^2$.
- (q) Find the average value of $f(x, y) = x^2y$ over $R = [0, 2] \times [0, 1]$.

22. Reversing Order of Integration (Double Integrals)

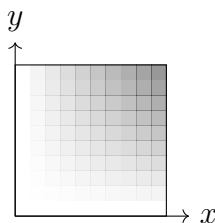
- (a) [2D] Evaluate $\int_0^4 \int_{\sqrt{y}}^2 \frac{y \cos(x^5)}{x} dx dy$.
- (b) Evaluate $\int_0^1 \int_{x^2}^1 x^3 \sin(y^3) dy dx$.
- (c) Reverse the order and evaluate $\int_0^1 \int_{\arcsin y}^{\pi/2} \cos(x) \sqrt{1 + \cos^2 x} dx dy$.
- (d) Find the volume of the solid in the first octant bounded by $z = 1 - x^2$, $y = 1 - x^2$, $x = 0$, $y = 0$, $z = 0$ by setting up the integral in two different orders.
- (e) Evaluate $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$.
- (f) Reverse the order of integration for $\int_0^2 \int_{e^y}^{e^2} f(x, y) dx dy$.
- (g) Sketch the region and reverse the order for $\int_0^1 \int_0^{x^2} f(x, y) dy dx$.

23. Changing Order in Triple Integrals

- (a) [3D] Rewrite the integral $\int_0^1 \int_0^{1-x^2} \int_0^{1-x} f(x, y, z) dz dy dx$ in the order $dy dz dx$.
- (b) Consider the solid bounded by $y = x^2$, $z = 0$, and $y + z = 1$. Set up the triple integral $\iiint_E f(x, y, z) dV$ in all six possible orders.
- (c) Evaluate $\int_0^1 \int_{\sqrt[3]{z}}^1 \int_0^{\ln 3} \frac{\pi e^{2x} \sin(\pi y^2)}{y^2} dx dy dz$.
- (d) Rewrite the integral $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z) dz dy dx$ in the order $dx dy dz$.
- (e) Let E be the region in the first octant bounded by the plane $2x + 3y + z = 6$. Write $\iiint_E f(x, y, z) dV$ as an iterated integral in the order $dz dx dy$.
- (f) Rewrite $\int_0^1 \int_x^1 \int_0^{1-y} f(x, y, z) dz dy dx$ in the order $dy dz dx$.
- (g) Sketch the solid and rewrite $\int_0^1 \int_0^{1-x} \int_0^2 dz dy dx$ in the order $dy dx dz$.

24. Area, Mass, and Moments (2D and 3D)

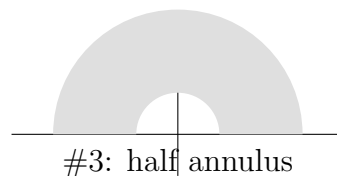
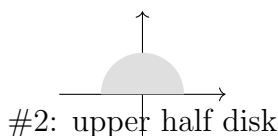
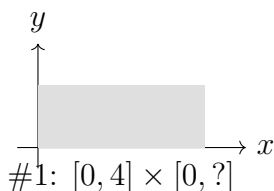
- (a) [2D] Find the mass of a lamina in the shape of the region bounded by $y = e^x$, $y = 0$, $x = 0$, $x = 1$ if the density is $\rho(x, y) = y$.
- (b) Find the center of mass $(\bar{x}, \bar{y})$ of a semicircular plate $x^2 + y^2 \leq a^2$, $y \geq 0$ with constant density ρ .
- (c) Let E be the solid in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$) bounded by the sphere $\rho = 2$ (i.e., $x^2 + y^2 + z^2 = 4$) with constant density $\delta = 1$.
- Set up and evaluate the moment $M_{xz} = \iiint_E y \, dV$ with respect to the xz -plane.
 - Set up and evaluate the moment $M_{xy} = \iiint_E z \, dV$ with respect to the xy -plane.
 - Find the z -coordinate of the center of mass $\bar{z} = M_{xy}/m$, where m is the mass of the solid.
- (d) A solid occupies the region bounded above by the paraboloid $z = 4 - x^2 - y^2$ and below by the xy -plane, with density $\delta(x, y, z) = z$.
- Compute the mass $m = \iiint_E \delta \, dV$ using cylindrical coordinates.
 - Compute the moment $M_{xy} = \iiint_E z \delta \, dV$ and find $\bar{z}$.
- (e) (Stewart §15.4 #2) Electric charge is distributed over the disk $x^2 + y^2 \leq 1$ so that the charge density at (x, y) is $\sigma(x, y) = \sqrt{x^2 + y^2}$ (in coulombs per square meter). Find the total charge.
- (f) (Stewart §15.4 #4) The figure shows a lamina on the unit square shaded according to density $\rho(x, y) = xy$. Estimate the location of the center of mass, then compute it exactly.



- (g) (Stewart §15.4 #7) Let D be the triangle with vertices $(0, 0)$, $(2, 1)$, $(0, 3)$ and density $\rho(x, y) = x + y$. Find the mass and the center of mass.
- (h) Find the centroid of the region bounded by $y = x$ and $y = x^2$.

25. Polar Coordinates

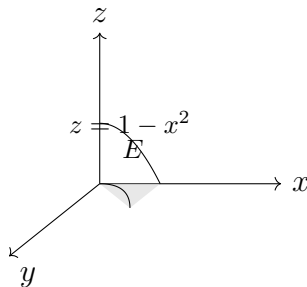
- (a) [2D] (Stewart §15.3 #1–4) For each region shown below, decide whether polar or rectangular coordinates are more appropriate, and write $\iint_R f(x, y) \, dA$ as an iterated integral.



- (b) (Stewart §15.3 #7) Sketch the region and evaluate $\int_{\pi/4}^{3\pi/4} \int_1^2 r \, dr \, d\theta$.
- (c) (Stewart §15.3 #8) Sketch the region and evaluate $\int_0^{\pi/2} \int_0^{2\sin\theta} r \, dr \, d\theta$.
- (d) (Stewart §15.3 #13) Evaluate $\iint_D e^{-(x^2+y^2)} \, dA$, where D is the region bounded by the semicircle $x = \sqrt{4-y^2}$ and the y -axis.
- (e) Set up and evaluate an integral in polar coordinates to find the area of one loop of the rose $r = \cos(3\theta)$.
- (f) Evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{3/2} \, dy \, dx$ by converting to polar coordinates.
- (g) Find the area of the region that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 3\cos\theta$.
- (h) Evaluate $\iint_D (x^2 + y^2) \, dA$ where D is the disk $x^2 + y^2 \leq 4$.
- (i) Find the area inside the cardioid $r = 1 + \cos\theta$.

26. Triple Integrals in Rectangular and Cylindrical

- (a) [3D] Evaluate $\iiint_E xyz \, dV$, where E is the rectangular box $[0, 1] \times [0, 2] \times [0, 3]$.
- (b) (Stewart §15.6 #9) Let E be the solid bounded above by $z = 1 - x^2$, below by the plane $z = 0$, and on the sides by the plane $y = 0$ and the surface $x = \sqrt{y}$ (see figure). For $f(x, y, z) = x$, do two things. (a) Express $\iiint_E f \, dV$ as an iterated integral. (b) Evaluate it.



- (c) (Stewart §15.6 #10) For $f(x, y, z) = xy$ and E bounded by $y + z = 2$, $z = 4 - y^2$, $y = x$, $x = 0$, write $\iiint_E f \, dV$ as an iterated integral and evaluate it.
- (d) Find the volume of the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.
- (e) (Stewart §15.6 #23, adapted) Evaluate $\iiint_T y^2 \, dV$ where T is the solid tetrahedron with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(0, 2, 0)$, $(0, 0, 2)$.
- (f) (Stewart §15.6 #32, adapted) Sketch the solid whose volume is given by

$$\int_0^2 \int_0^{2-y} \int_0^{4-y^2} dx \, dz \, dy.$$

- (g) Use cylindrical coordinates to find the volume of the solid that lies within the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$.
- (h) (Stewart §15.7 #11) Sketch the solid described by $r^2 \leq z \leq 8 - r^2$.
- (i) (Stewart §15.7 #12) Sketch the solid described by $0 \leq \theta \leq \pi/2$, $r \leq z \leq 2$.
- (j) (Stewart §15.7 #18) Sketch the solid whose volume is given by

$$\int_0^2 \int_0^{2\pi} \int_0^r r \, dz \, d\theta \, dr,$$

and evaluate the integral.

- (k) (Stewart §15.7 #19) Use cylindrical coordinates to evaluate $\iiint_E \sqrt{x^2 + y^2} \, dV$ where E is the region inside the cylinder $x^2 + y^2 = 16$ and between $z = -5$ and $z = 4$.
- (l) (Stewart §15.7 #20) Evaluate $\iiint_E z \, dV$ where E is enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
- (m) Evaluate $\iiint_E (x^2 + y^2) \, dV$, where E is the region bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = -1$ and $z = 2$.
- (n) (Stewart §15.7 #26) Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.

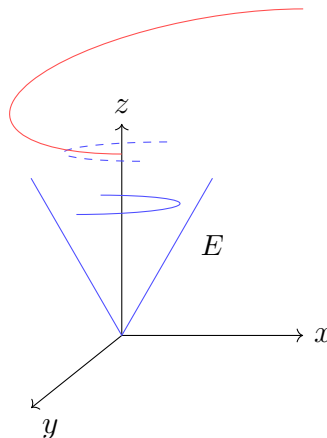
- (o) (Stewart §15.7 #32) Evaluate by changing to cylindrical coordinates.

$$\int_0^2 \int_0^{\sqrt{4-y^2}} \int_{-2}^2 xz \, dz \, dx \, dy.$$

- (p) Find the volume of the solid bounded by the cylinders $x^2 + y^2 = 1$ and $x^2 + z^2 = 1$ (the Steinmetz solid)
- (q) Find the volume of the solid bounded below by $z = x^2 + y^2$ and above by $z = 8 - x^2 - y^2$.
- (r) Find the mass of the cylinder $x^2 + y^2 \leq 1$, $0 \leq z \leq 2$, with density $\rho = z$.

27. Triple Integrals in Spherical Coordinates

- (a) [3D] Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the unit ball.
- (b) (Stewart §15.8 #19) Set up the triple integral of an arbitrary continuous function $f(x, y, z)$ in cylindrical or spherical coordinates over the solid shown below, a cone of half-angle $\pi/3$ inside a sphere of radius 3 centered at the origin, the ice cream cone.



- (c) (Stewart §15.8 #24) Evaluate $\iiint_E y^2 z^2 dV$, where E lies above the cone $\phi = \pi/3$ and below the sphere $\rho = 1$.
- (d) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.
- (e) (Stewart §15.8 #27) Evaluate $\iiint_E x e^{x^2+y^2+z^2} dV$, where E is the portion of the unit ball $x^2 + y^2 + z^2 \leq 1$ that lies in the first octant.
- (f) (Stewart §15.8 #30) Find the average distance from a point in a ball of radius a to its center.
- (g) Set up the integral to find the mass of a spherical shell $1 \leq \rho \leq 2$ if the density is $\delta(\rho, \phi, \theta) = \rho$.
- (h) Find the volume of the “ice cream cone” solid cut from the sphere $\rho \leq a$ by the cone $\phi = \alpha$.
- (i) Let E be the region in the first octant ($x \geq 0, y \geq 0, z \geq 0$) between the spheres of radius 1 and radius 3 centered at the origin.
- Write the bounds for ρ , ϕ , and θ in spherical coordinates.
 - Evaluate $\iiint_E z dV$.
- (j) Use spherical coordinates to evaluate $\iiint_E (x^2 + y^2 + z^2) dV$ where E is the solid in the first octant bounded by the sphere $x^2 + y^2 + z^2 = 9$ and the coordinate planes.
- (k) (Stewart §15.8 #44) Evaluate by changing to spherical coordinates.

$$\int_{-a}^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} \int_{-\sqrt{a^2-x^2-y^2}}^{\sqrt{a^2-x^2-y^2}} (x^2 z + y^2 z + z^3) dz dx dy.$$

- (l) Find the volume of a ball of radius a using spherical coordinates.

28. Coordinate Transformations (Jacobians)

- (a) [2D] Evaluate $\iint_R (x+y)e^{x-y} dA$ where R is the diamond bounded by $x+y=1, x+y=3, x-y=-1, x-y=1$ using $u=x+y$ and $v=x-y$.
- (b) Use the transformation $x=au, y=bv$ to find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
- (c) Evaluate $\iint_R \cos\left(\frac{y-x}{y+x}\right) dA$ where R is the triangle with vertices $(0,0), (1,0), (0,1)$.
- (d) Compute the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$ for $x=u^2-v^2, y=2uv$.
- (e) Confirm that the polar change of variables gives $\frac{\partial(x,y)}{\partial(r,\theta)} = r$.

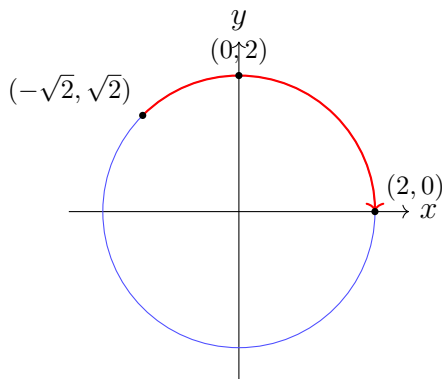
Vector Calculus (Part A)

29. Vector Fields

- (a) [2D] The dome's gradient field is $\vec{F} = \nabla f = \langle -2x, -2y \rangle$, where $f = 1 - x^2 - y^2$. Draw $\vec{F}$ at $(\pm 1, 0)$, $(0, \pm 1)$, and $(\pm 1, \pm 1)$, each arrow scaled down by half. Which way does every arrow point, and how does that match water running off the dome?
- (b) [2D] On the same axes sketch a few level curves of f (circles) and check that each arrow of $\vec{F}$ crosses its circle at a right angle, longest where the circles crowd together. This is the gradient picture from the last chapter, now drawn as a field.
- (c) [2D] Contrast the swirl $\vec{G} = \langle -y, x \rangle$. Draw it at the same points. Walk a small loop and watch the height fail to return, then say in one line why no surface has $\vec{G}$ as its gradient.
- (d) (Stewart §16.1 #2) Sketch the vector field $\vec{F}(x, y) = 2\hat{\mathbf{i}} - \hat{\mathbf{j}}$.
- (e) (Stewart §16.1 #4) Sketch $\vec{F}(x, y) = x\hat{\mathbf{i}} + \frac{1}{2}y\hat{\mathbf{j}}$.
- (f) (Stewart §16.1 #8) Sketch $\vec{F}(x, y) = \frac{y\hat{\mathbf{i}} - x\hat{\mathbf{j}}}{\sqrt{x^2 + y^2}}$.
- (g) (Stewart §16.1 #10) Sketch the 3D vector field $\vec{F}(x, y, z) = z\hat{\mathbf{i}}$.
- (h) (Stewart §16.1 #13–18, matching) Match each formula to its plot in Stewart §16.1 and explain.
- $\vec{F}(x, y) = \langle x, -y \rangle$
 - $\vec{F}(x, y) = \langle y, y + 2 \rangle$
 - $\vec{F}(x, y) = \langle y, -x \rangle$
 - $\vec{F}(x, y) = \langle \sin y, \cos x \rangle$
- (i) (Stewart §16.1 #19–22, matching) Match each 3D vector field to its plot in Stewart §16.1.
- $\vec{F} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$
 - $\vec{F} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + z\hat{\mathbf{k}}$
 - $\vec{F} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$
 - $\vec{F} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$
- (j) (Stewart §16.1 #31–34, matching) Match each function f to the plot of its gradient field ∇f in Stewart §16.1.
- $f(x, y) = x^2 + y^2$
 - $f(x, y) = x(x + y)$
 - $f(x, y) = (x + y)^2$
 - $f(x, y) = \sin \sqrt{x^2 + y^2}$
- (k) Sketch the vector fields $\vec{F}(x, y) = \langle x, y \rangle$ and $\vec{F}(x, y) = \langle x, -y \rangle$.
- (l) Find the gradient field ∇f for $f(x, y) = x^2 + y^2$ and for $f(x, y) = x^2 - y^2$, and sketch each.
- (m) Determine whether $\vec{F} = \langle y, x \rangle$ is a gradient field, and if so find a function f with $\nabla f = \vec{F}$.

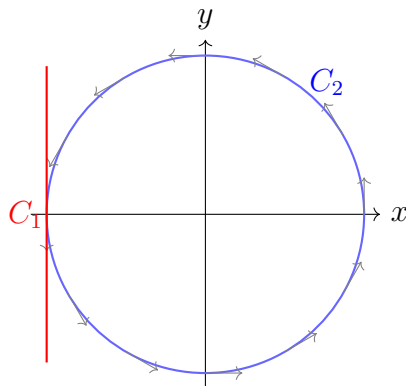
30. Line Integrals (Scalar and Vector)

- (a) [1D IN 3D] Evaluate the line integral $\int_C y \, ds$, where C is the right half of the circle $x^2 + y^2 = 1$ from $(0, -1)$ to $(0, 1)$.
- (b) (Stewart §16.2 #1) Evaluate $\int_C y \, ds$, where C is given by $x = t^3$, $y = 2t$, $0 \leq t \leq 3/2$.
- (c) (Stewart §16.2 #3) Evaluate $\int_C xy^4 \, ds$ where C is the right half of the circle $x^2 + y^2 = 16$.
- (d) (Stewart §16.2 #9) Evaluate $\int_C x^2 y \, ds$ where C is given by $x = \cos t$, $y = \sin t$, $z = t$, $0 \leq t \leq \pi/2$.
- (e) (Stewart §16.2 #11) Evaluate $\int_C x e^{yz} \, ds$ where C is the line segment from $(0, 0, 0)$ to $(1, 2, 3)$.
- (f) (Stewart §16.2 #12) Evaluate $\int_C (x^2 + y^2 + z^2) \, ds$ where C is given by $x = t$, $y = \cos 2t$, $z = \sin 2t$, $0 \leq t \leq 2\pi$.
- (g) Find the work done by the force field $\vec{F}(x, y) = \langle x^2, -xy \rangle$ along the quarter-circle $\vec{r}(t) = \langle \cos t, \sin t \rangle$ for $0 \leq t \leq \pi/2$.
- (h) (Stewart §16.2 #5) Evaluate $\int_C (x^2 y + \sin x) \, dy$ where C is the arc of the parabola $y = x^2$ from $(0, 0)$ to (π, π^2) .
- (i) (Stewart §16.2 #8) Evaluate $\int_C x^2 \, dx + y^2 \, dy$ where C is the arc of $x^2 + y^2 = 4$ from $(-\sqrt{2}, \sqrt{2})$ to $(2, 0)$ traversed through $(0, 2)$ (see figure).



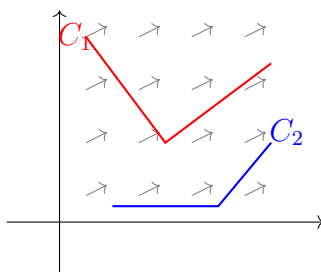
- (j) Evaluate $\int_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = \langle z, x, y \rangle$ and C is the helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \leq t \leq \pi$.
- (k) (Stewart §16.2 #15) Evaluate $\int_C z \, dx + xy \, dy + y^2 \, dz$ where C is given by $x = \sin t$, $y = \cos t$, $z = \tan t$, $-\pi/4 \leq t \leq \pi/4$.
- (l) Evaluate the scalar line integral $\int_C (x + yz) \, ds$ where C is the twisted cubic $\vec{r}(t) = \langle t, t^2, t^3 \rangle$ for $0 \leq t \leq 1$.
- (m) Evaluate $\int_C (x^2 + y^2) \, ds$ where C is the “bent” path from $(0, 0, 0)$ to $(1, 2, 3)$ consisting of three straight line segments parallel to the axes.
- (n) Find the work done by $\vec{F} = \langle \sin x, \cos y, xz \rangle$ along the curve $\vec{r}(t) = \langle t^3, -t^2, t \rangle$ from $t = 0$ to $t = 1$.
- (o) (Stewart §16.2 #21) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F}(x, y) = xy^2 \hat{i} - x^2 \hat{j}$ where $\vec{r}(t) = t^3 \hat{i} + t^2 \hat{j}$, $0 \leq t \leq 1$.

(p) (Stewart §16.2 #19) The vector field $\vec{F}$ is shown below.



(a) If C_1 is the vertical line segment from $(-3, -3)$ to $(-3, 3)$, decide if $\int_{C_1} \vec{F} \cdot d\vec{r}$ is positive, negative, or zero. (b) If C_2 is the counterclockwise circle of radius 3 centered at the origin, decide the sign of $\int_{C_2} \vec{F} \cdot d\vec{r}$.

(q) (Stewart §16.2 #20) The figure below shows another field $\vec{F}$ and two curves C_1, C_2 . Are the line integrals positive, negative, or zero? Explain.



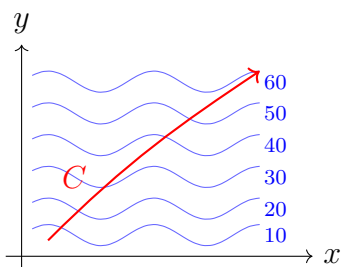
(r) Evaluate $\int_C xy \, ds$ where C is the segment from $(0, 0)$ to $(1, 2)$.

(s) Evaluate $\int_C y \, dx + x \, dy$ along $\vec{r}(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.

(t) Find the work done by $\vec{F} = \langle x + y, x - y \rangle$ along the segment from $(0, 0)$ to $(2, 3)$.

31. Conservative Fields and Potential Functions

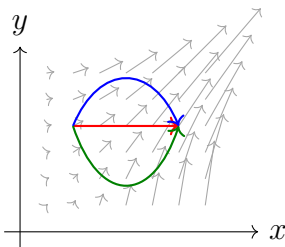
- (a) [2D] The dome height $f = 1 - x^2 - y^2$ is a potential for its gradient field $\vec{F} = \langle -2x, -2y \rangle$, so $\vec{F}$ is conservative. Argue the work along any curve from $(1, 0)$ to $(0, 0)$ is just the height gained, $f(0, 0) - f(1, 0)$, whatever the path. Sketch two different paths and confirm they agree.
- (b) [2D] (Stewart §16.3 #1) The figure shows a curve C and a contour map of a function f whose gradient is continuous. The curve C starts on the level $f = 10$ and ends on the level $f = 60$. Use the picture to find $\int_C \nabla f \cdot d\vec{r}$.



- (c) (Stewart §16.3 #2) A table of values of a function f with continuous gradient is given (selected values shown). Find $\int_C \nabla f \cdot d\vec{r}$, where C has parametric equations $x = t^2 + 1$, $y = t^3 + t$, $0 \leq t \leq 1$.

$x \backslash y$	0	1	2
0	1	6	4
1	3	5	7
2	8	2	9

- (d) (Stewart §16.3 #4) Determine whether $\vec{F}(x, y) = (y - 2x)\hat{\mathbf{i}} + 2xy\hat{\mathbf{j}}$ is conservative, and if so find f with $\vec{F} = \nabla f$.
- (e) (Stewart §16.3 #8) Determine whether $\vec{F}(x, y) = (2xy + y^{-2})\hat{\mathbf{i}} + (x^2 - 2xy^{-3})\hat{\mathbf{j}}$ (with $y > 0$) is conservative.
- (f) Show that $\vec{F}(x, y, z) = \langle yz, xz, xy \rangle$ is conservative and find f such that $\nabla f = \vec{F}$.
- (g) Use the Fundamental Theorem for Line Integrals to evaluate $\int_C \nabla f \cdot d\vec{r}$ where $f(x, y) = \sin(x) + x^2y$ and C is any path from $(0, 0)$ to $(\pi, 2)$.
- (h) (Stewart §16.3 #11) The figure shows the vector field $\vec{F}(x, y) = \langle 2xy, x^2 \rangle$ and three curves C_1, C_2, C_3 that each start at $(1, 2)$ and end at $(3, 2)$.

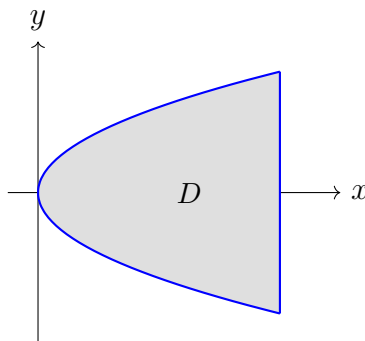


- (a) Explain why $\int_{C_i} \vec{F} \cdot d\vec{r}$ has the same value for all three. (b) Compute this common value.
- (i) (Stewart §16.3 #12) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F}(x, y) = 2xy\hat{\mathbf{i}} + (x^2 + \sin y)\hat{\mathbf{j}}$ along (a) any path from $(0, 0)$ to $(2, \pi/2)$ and (b) any path from $(0, 0)$ to $(1, 1)$ along $y = \sin(\pi x/2)$.

- (j) (Stewart §16.3 #13) Let $\vec{F}(x, y) = (3x^2 + y^2)\hat{\mathbf{i}} + 2xy\hat{\mathbf{j}}$ and let C be the upper half of the circle of radius 2 centered at the origin, from $(-2, 0)$ to $(2, 0)$. (a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ directly. (b) Show $\vec{F}$ is conservative and find f . (c) Re-evaluate using the FTC. (d) Replace C by a simpler path with the same endpoints and re-evaluate.
- (k) (Stewart §16.3 #14) Let $\vec{F}(x, y) = \langle \sin y + e^x, x \cos y \rangle$ and C given by $x = t$, $y = t(3 - t)$, $0 \leq t \leq 3$. (a) Show $\vec{F}$ is conservative, find f . (b) Use FTC. (c) Replace C by a line segment with the same endpoints, then re-evaluate.
- (l) Find the work done by the field $\vec{F} = \langle 2xe^y, x^2e^y + \cos y \rangle$ along the path $y = \sin x$ from $(0, 0)$ to $(\pi, 0)$.
- (m) (Stewart §16.3 #18) Find f with $\nabla f = \vec{F}$ for $\vec{F} = (3 + 2xy^2)\hat{\mathbf{i}} + 2x^2y\hat{\mathbf{j}}$, and use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the arc of $y = 1/x$ from $(1, 1)$ to $(4, 1/4)$.
- (n) (Stewart §16.3 #20) Find f such that $\vec{F} = \nabla f$ for $\vec{F} = (1 + xy)e^{xy}\hat{\mathbf{i}} + x^2e^{xy}\hat{\mathbf{j}}$, then evaluate $\int_C \vec{F} \cdot d\vec{r}$ on $\vec{r}(t) = \cos t\hat{\mathbf{i}} + 2\sin t\hat{\mathbf{j}}$, $0 \leq t \leq \pi/2$.
- (o) Determine if $\vec{F} = \langle e^y, xe^y + e^z, ye^z \rangle$ is conservative. If so, find f .
- (p) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = \langle e^y, xe^y + e^z, ye^z \rangle$ along any path from $(0, 0, 0)$ to $(1, 1, 1)$.
- (q) Show $\vec{F} = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative and find a potential f .
- (r) Using the potential from the previous part, evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, 2)$.
- (s) Determine whether $\vec{F} = \langle y^2, 2xy + e^{3z}, 3ye^{3z} \rangle$ is conservative, and if so find a potential.
- (t) Is $\vec{F} = \frac{\langle -y, x \rangle}{x^2 + y^2}$ conservative on the plane with the origin removed? Test $P_y = Q_x$ and explain the role of the hole.

32. Green's Theorem

- (a) [2D] (Stewart §16.4 #1) Evaluate $\oint_C y^2 dx + x^2 y dy$ where C is the rectangle with vertices $(0, 0)$, $(5, 0)$, $(5, 4)$, $(0, 4)$, by two methods, first directly and then using Green's Theorem.
- (b) (Stewart §16.4 #2) Evaluate $\oint_C y dx - x dy$ where C is the circle of radius 4 centered at the origin, by (a) direct computation and (b) Green's Theorem.
- (c) Use Green's Theorem to evaluate $\oint_C (y^2 dx + 3xy dy)$, where C is the boundary of the upper semi-annulus $1 \leq x^2 + y^2 \leq 4, y \geq 0$.
- (d) (Stewart §16.4 #6) Use Green's Theorem to evaluate $\oint_C \ln(xy) dx + (y/x) dy$ where C is the rectangle with vertices $(1, 1)$, $(1, 4)$, $(2, 4)$, $(2, 1)$.
- (e) (Stewart §16.4 #8) Use Green's Theorem on $\oint_C (x^2 + y^2) dx + (x^2 - y^2) dy$ where C is the triangle with vertices $(0, 0)$, $(2, 1)$, $(0, 1)$.
- (f) Use Green's Theorem to find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using the formula $A = \frac{1}{2} \oint_C (x dy - y dx)$.
- (g) Evaluate $\oint_C (xy + y^2) dx + (x^2) dy$ where C is the boundary of the region between $y = x$ and $y = x^2$.
- (h) (Stewart §16.4 #12) Use Green's Theorem to evaluate $\oint_C (1 - y^3) dx + (x^3 + e^{y^2}) dy$ where C is the boundary of the region between $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$ (positively oriented).
- (i) (Stewart §16.4 #14) Use Green's Theorem to compute $\oint_C (x^{2/3} + y^3) dx + (y^{4/3} - x^3) dy$ where C is the boundary of the region bounded by $x = y^2$ and $x = 4$, traversed positively (see figure).



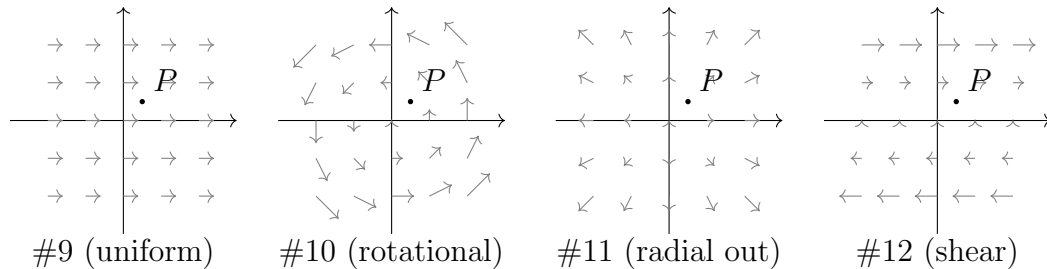
- (j) Use Green's Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \langle e^x - y^3, \cos y + x^3 \rangle$ and C is the unit circle.
- (k) Use Green's Theorem to find the work done by $\vec{F} = \langle y + e^{\sqrt{x}}, 2x + \cos(y^2) \rangle$ on a particle moving once counterclockwise around the unit circle.
- (l) Evaluate $\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$ where C is a circle of radius 1 centered at $(0, 0)$. Why does Green's Theorem require care here?
- (m) Use Green's Theorem to evaluate $\oint_C (y^2 dx + x^2 dy)$ around the unit square, counterclockwise.
- (n) Use Green's Theorem to evaluate $\oint_C (xy dx + x^2 dy)$ around the triangle $(0, 0)$, $(1, 0)$, $(1, 1)$.

Pre-Final Material

Vector Calculus (Part B)

33. Curl and Divergence

- (a) [2D] Return to the dome's two fields. For the gradient field $\vec{F} = \langle -2x, -2y \rangle$ compute $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, and read the signs as a sink at the peak with no spin. For the swirl $\vec{G} = \langle -y, x \rangle$ compute both and read a pure spin with no net outflow. Sketch a tiny paddle wheel in each field at $(1, 0)$.
- (b) [3D] (Stewart §16.5 #1) Find (a) the curl and (b) the divergence of $\vec{F}(x, y, z) = xy^2z^2 \hat{i} + x^2yz^2 \hat{j} + x^2y^2z \hat{k}$.
- (c) Find the curl and divergence of the vector field $\vec{F}(x, y, z) = \langle xz, xyz, -y^2 \rangle$.
- (d) (Stewart §16.5 #9–12, figure-based) For each planar vector field shown, independent of z with zero z -component, determine the following at the marked point P . (a) Is $\text{div } \vec{F}$ positive, negative, or zero? (b) Is $\text{curl } \vec{F}$ nonzero, and if so does it point in $+\hat{k}$ or $-\hat{k}$ at P ?



- (e) (Stewart §16.5 #13) (a) Verify Formula 3 (i.e. $\text{curl}(\nabla f) = \vec{0}$) for $f(x, y, z) = \sin(xyz)$.
 (b) Verify Formula 11 (i.e. $\text{div}(\text{curl } \vec{F}) = 0$) for $\vec{F} = xyz^2 \hat{i} + x^2yz^3 \hat{j} + y^2 \hat{k}$.
- (f) (Stewart §16.5 #14) Let f be a scalar field and $\vec{F}$ a vector field. State whether each expression is meaningful, and if so say whether it is a scalar field or a vector field. (a) $\text{curl } f$, (b) $\text{grad } f$, (c) $\text{div } \vec{F}$, (d) $\text{curl}(\text{grad } f)$, (e) $\text{grad } \vec{F}$, (f) $\text{grad}(\text{div } \vec{F})$, (g) $\text{div}(\text{grad } f)$, (h) $\text{grad}(\text{div } f)$, (i) $\text{curl}(\text{curl } \vec{F})$.
- (g) (Stewart §16.5 #15) Determine whether $\vec{F}(x, y, z) = \langle 2xy^3z^2, 3x^2y^2z^2, 2x^2y^3z \rangle$ is conservative, and if so find f such that $\vec{F} = \nabla f$.
- (h) (Stewart §16.5 #16) Determine whether $\vec{F}(x, y, z) = \langle yz, xz + y, xy - x \rangle$ is conservative, and if so find f .
- (i) Show that any vector field of the form $\vec{F} = \langle f(x), g(y), h(z) \rangle$ has $\text{curl } \vec{F} = \vec{0}$.
- (j) Prove that $\text{div}(\text{curl } \vec{F}) = 0$ for any smooth vector field $\vec{F}$.
- (k) If $f(x, y, z)$ has continuous second partial derivatives, prove that $\text{curl}(\nabla f) = \vec{0}$.
- (l) (Stewart §16.5 #21) Is there a vector field $\vec{G}$ on $\mathbb{R}^3$ such that $\text{curl } \vec{G} = \langle x \sin y, \cos y, z - xy \rangle$? Explain.
- (m) Find the curl and divergence of $\vec{F} = \langle x^2, y^2, z^2 \rangle$ and of $\vec{G} = \langle e^x \sin y, e^x \cos y, z \rangle$.

34. Parametric Surfaces and Surface Area

- (a) [2D IN 3D] Find a parameterization $\vec{r}(u, v)$ for the portion of the plane $x + y + z = 1$ that lies in the first octant.
- (b) Find the surface area of the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 4$.
- (c) Find the area of the part of the plane $2x + 5y + z = 10$ that lies inside the cylinder $x^2 + y^2 = 9$.
- (d) Find the surface area of the part of the sphere $x^2 + y^2 + z^2 = a^2$ that lies inside the cylinder $x^2 + y^2 = ax$.
- (e) Parameterize the cylinder $x^2 + y^2 = 4$, $0 \leq z \leq 3$, and the sphere of radius 2.

35. Surface Integrals and Flux

- (a) [2D IN 3D] Evaluate $\iint_S x^2 dS$, where S is the unit sphere $x^2 + y^2 + z^2 = 1$.
- (b) Find the flux of $\vec{F} = \langle z, y, x \rangle$ across the unit square in the xy -plane ($0 \leq x \leq 1, 0 \leq y \leq 1$) with upward orientation.
- (c) Find the flux of $\vec{F} = \langle x, y, z^4 \rangle$ across the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 1$.
- (d) Evaluate $\iint_S z dS$ where S is the part of the plane $z = 1 + 2x + 3y$ over $[0, 1] \times [0, 1]$.
- (e) Find the flux of $\vec{F} = \langle x, y, z \rangle$ outward through the unit sphere.

36. Stokes' Theorem

- (a) [2D IN 3D] Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = \langle -y^2, x, z^2 \rangle$ and C is the circle $x^2 + y^2 = 1$ in the plane $z = 2$.
- (b) Use Stokes' Theorem to evaluate $\iint_S (\text{curl } \vec{F}) \cdot d\vec{S}$, where $\vec{F} = \langle xz, yz, xy \rangle$ and S is the upper hemisphere $x^2 + y^2 + z^2 = 1, z \geq 0$.
- (c) Verify Stokes' Theorem for $\vec{F} = \langle y, z, x \rangle$ and the surface $z = 1 - x^2 - y^2, z \geq 0$.

37. The Divergence Theorem

- (a) [3D] Use the Divergence Theorem to find the flux of $\vec{F} = \langle z, y, x \rangle$ across the surface of the unit cube.
- (b) Find the flux of $\vec{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = a^2$.
- (c) Use the Divergence Theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$ for $\vec{F} = \langle xy, yz, zx \rangle$ over the boundary of the cylinder $x^2 + y^2 \leq 1, 0 \leq z \leq 1$.
- (d) Calculate the flux of $\vec{F} = \langle z, y, x \rangle$ across the top half of the sphere $x^2 + y^2 + z^2 = 1, z \geq 0$, by closing the surface and using the Divergence Theorem.
- (e) Use the Divergence Theorem to find the flux of $\vec{F} = \langle x, y, z \rangle$ out of the unit ball.
- (f) Use the Divergence Theorem to find the flux of $\vec{F} = \langle x^2, y^2, z^2 \rangle$ out of the box $[0, 1]^3$.

Lagrange Multipliers

38. Constrained Optimization (One Constraint)

- (a) [1D IN 2D] Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = x^2 + y^2$ subject to the constraint $xy = 1$.
- (b) Find the dimensions of a rectangular box with largest volume such that the sum of the lengths of its 12 edges is a constant L .
- (c) Find the maximum and minimum values of $f(x, y) = 3x + y$ on the circle $x^2 + y^2 = 10$.
- (d) Find the point on the plane $x - y + z = 4$ that is closest to the point $(1, 2, 3)$. Minimize the square of the distance $d^2 = (x - 1)^2 + (y - 2)^2 + (z - 3)^2$.
- (e) A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.
- (f) Find the maximum and minimum values of $f(x, y) = e^{xy}$ subject to the constraint $x^3 + y^3 = 16$ for $x, y > 0$.
- (g) Use Lagrange multipliers to find the extrema of $f(x, y) = xy$ subject to $x^2 + y^2 = 8$.

39. Multiple Constraints and Geometry

- (a) [1D IN 3D] Find the maximum value of $f(x, y, z) = x + 2y + 3z$ on the curve of intersection of the plane $x - y + z = 1$ and the cylinder $x^2 + y^2 = 1$.
- (b) **Challenge, two constraints.** Use the method of Lagrange multipliers with two constraints, $\nabla f = \lambda \nabla g + \mu \nabla h$, to find the highest and lowest points on the ellipse formed by the intersection of the cone $z^2 = x^2 + y^2$ and the plane $x + 2z = 4$.
- (c) **Challenge.** Prove that the product of the sines of the angles of a triangle, $f(\alpha, \beta, \gamma) = \sin \alpha \sin \beta \sin \gamma$, is maximized when the triangle is equilateral, under the constraint $\alpha + \beta + \gamma = \pi$.
- (d) Find the point on the intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 1$ nearest the origin.

Connections to Future Math

40. Probability and Multiple Integrals

- (a) A joint density on the unit square is $f(x, y) = x + y$ on $0 \leq x \leq 1$, $0 \leq y \leq 1$. Verify that $\iint f \, dA = 1$.
- (b) For that density find $P(X \leq \frac{1}{2}, Y \leq \frac{1}{2}) = \int_0^{1/2} \int_0^{1/2} (x + y) \, dy \, dx$.
- (c) Find the marginal density $f_X(x) = \int_0^1 (x + y) \, dy$ and check that $\int_0^1 f_X(x) \, dx = 1$.
- (d) Find the expected value $E[X] = \iint x f \, dA$, and notice it is the $\bar{x}$ of the density read as a lamina.
- (e) Let $f(x, y) = c e^{-(x+y)}$ on the first quadrant. Find c so the total is one, then find $P(X + Y \leq 1)$.

41. Fourier Series and Orthogonality

- (a) With $\langle f, g \rangle = \int_{-\pi}^{\pi} f g \, dx$, show that $\sin x$ and $\cos x$ are orthogonal.
- (b) Show that $\langle \sin(mx), \sin(nx) \rangle = 0$ for integers $m \neq n$, and equals π when $m = n$.
- (c) Find $b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin x \, dx$, the projection of $f(x) = x$ onto $\sin x$.
- (d) For the square wave $f(x) = 1$ on $(0, \pi)$ and $f(x) = -1$ on $(-\pi, 0)$, find $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) \, dx$, and note that only odd n survive.

42. Linear Algebra, Transformations and Systems

- (a) [2D] Apply $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ to the four corners of the unit square and sketch the image. What is the area scale, and how does it match $|\det A|$?
- (b) Apply the shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ to the unit square and sketch the image. Find $\det A$ and confirm the area is unchanged.
- (c) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ compute $A\langle 1, 1 \rangle$, then solve $A\vec{x} = \langle 5, 11 \rangle$.
- (d) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ find every $\vec{x}$ with $A\vec{x} = \vec{0}$ and describe the set. Compute $\det A$ and explain why a zero determinant forces a whole line of solutions.
- (e) A rotation by θ has matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Show $\det = 1$ and that it preserves the length of every vector, so it turns the plane without stretching it.

43. Differential Geometry, Manifolds and Implicit Surfaces

- (a) [2D IN 3D] The sphere $x^2 + y^2 + z^2 = 1$ is the level set $F = 0$ of $F(x, y, z) = x^2 + y^2 + z^2 - 1$. It is cut out of $\mathbb{R}^n$, $n = 3$, by $m = 1$ equation, so it is a 2-dimensional surface in $\mathbb{R}^3$, since $n - m = 3 - 1 = 2$. Find ∇F at $(0, 0, 1)$ and write the tangent plane there.
- (b) The circle $x^2 + y^2 = 1$, $z = 0$ is cut by two equations, so $n - m = 1$. Find both gradients at $(1, 0, 0)$ and check the tangent line is perpendicular to each.
- (c) Show the cone $x^2 + y^2 - z^2 = 0$ is not smooth at the origin by computing ∇F there, where $F = x^2 + y^2 - z^2$, and finding that it vanishes.
- (d) Parameterize the unit sphere by $\vec{r}(\phi, \theta) = \langle \sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi \rangle$, one chart (a parametrization of part of the sphere) in its atlas of charts, and compute $\vec{r}_\phi \times \vec{r}_\theta$ at $\phi = \frac{\pi}{2}$, $\theta = 0$. Confirm it is normal to the sphere.
- (e) Write $z = x^2 - y^2$ as $F = x^2 - y^2 - z = 0$ and find the tangent plane at $(1, 1, 0)$ two ways, from ∇F and from $z = f(a, b) + f_x \Delta x + f_y \Delta y$, and confirm they agree.

44. Variational Calculus, the Geometry of Paths and Energy

- (a) [1D IN 2D] The straight path from $(0, 0)$ to $(1, 1)$ is $y = x$. Perturb it to $y = x + \epsilon \sin(\pi x)$, which fixes the endpoints, and show that $\left. \frac{d}{d\epsilon} \right|_0 \int_0^1 \sqrt{1 + (y')^2} dx = 0$, so the line is a stationary path.
- (b) The Euler-Lagrange equation for $\int F(x, y, y') dx$ is $\frac{d}{dx} F_{y'} - F_y = 0$. For the arc length integrand $F = \sqrt{1 + (y')^2}$ show it gives $y'' = 0$, so straight lines are the plane geodesics (shortest paths).
- (c) For $F = \frac{1}{2}(y')^2$ write the Euler-Lagrange equation and solve it, reading $y'' = 0$ as motion at constant velocity.
- (d) A hanging chain minimizes $\int y \sqrt{1 + (y')^2} dx$. Since the integrand has no explicit x , the Beltrami identity $F - y' F_{y'} = C$ holds. Set up that equation, whose solution is the catenary $y = a \cosh(x/a)$. Setting it up is enough.

45. Differential Forms, the Universal Calculus

- (a) For the 0-form $f = x^2 y$ compute $df = f_x dx + f_y dy$ and match it to ∇f .
- (b) For the 1-form $\omega = -y dx + x dy$ the exterior derivative is $d\omega = (Q_x - P_y) dx \wedge dy$. Compute it and match it to the scalar curl of $\langle -y, x \rangle$.
- (c) Show $d(df) = 0$ for $f = x^2 y$, using $dx \wedge dx = 0$ and $dx \wedge dy = -dy \wedge dx$. This is the statement that the curl of a gradient is zero.
- (d) Write Green's Theorem as $\int_{\partial D} \omega = \int_D d\omega$ with $\omega = P dx + Q dy$, and identify ω and $d\omega$.
- (e) For the 2-form $\eta = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$ compute $d\eta = (P_x + Q_y + R_z) dx \wedge dy \wedge dz$ and match it to the divergence, so the Divergence Theorem is again $\int_{\partial \Omega} \eta = \int_{\Omega} d\eta$.

46. Complex Analysis and the Residue Theorem

- (a) Verify the Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ for $f(z) = z^2$, where $u = x^2 - y^2$ and $v = 2xy$.
- (b) Check that $u = x^2 - y^2$ is harmonic, $u_{xx} + u_{yy} = 0$, and find a conjugate v so that $u + iv$ is analytic.
- (c) Parameterize the unit circle by $z = e^{i\theta}$ and evaluate $\oint_C \frac{dz}{z}$ directly, obtaining $2\pi i$.
- (d) Evaluate $\oint_C \frac{dz}{z - a}$ when C is a circle that encloses a and when it does not, and state the pattern.
- (e) Use residues to evaluate $\oint_C \frac{e^z}{z - 2} dz$ on $|z| = 3$. Identify the pole at $z = 2$, read the residue e^2 , and conclude the integral is $2\pi i e^2$.

47. Physics, the Faraday 2-Form

- (a) Gauss's law equates the flux of $\vec{E}$ out of a closed surface with the enclosed charge over ϵ_0 . For a point charge $\vec{E} = \frac{q}{4\pi\epsilon_0} \frac{\hat{r}}{r^2}$, explain why the flux through every enclosing sphere is the same.
- (b) Faraday's law is $\text{curl } \vec{E} = -\partial_t \vec{B}$. For $\vec{B} = \langle 0, 0, t \rangle$ find $\text{curl } \vec{E}$ and one field $\vec{E}$ that produces it.
- (c) Since $\text{div } \vec{B} = 0$ there is a potential $\vec{A}$ with $\vec{B} = \text{curl } \vec{A}$. For the constant field $\vec{B} = \langle 0, 0, B \rangle$ verify that $\vec{A} = \frac{1}{2} \langle -By, Bx, 0 \rangle$ works. (This is the field form of the statement that the Faraday 2-form F , built from $\vec{E}$ and $\vec{B}$, satisfies $dF = 0$ and so has a potential.)